Olympiad Combinatorics

Learn it, then practise it: a method pack for BMO1

11 techniques · explanations and worked examples · 39 questions with full solutions

How to use this pack

Combinatorics at olympiad level is less about knowing formulas and more about recognising which *idea* a problem needs. This pack is built to teach those ideas from scratch, then make you use them.

Every section has two halves.

1. **Learn.** The idea is explained in plain language, with the reasoning behind every fact. Then come **worked examples**, solved step by step with commentary on *why* each step is taken, not just what it is. Coloured boxes mark the **key facts**, the **method** (when to reach for it and how), **why it works**, and the **common traps**. Each Learn part ends with a short **quick check**; the answers are printed upside down.
2. **Practise.** The questions. Each has its own page with room to work and a hint printed upside down at the bottom. Only turn it round when you've been stuck for a while. The model solution is on the next page, opening with the *key idea* in one or two sentences.

How to study a section. Read the Learn part with a pencil. Before you read the solution to a worked example, cover it and try it yourself for a few minutes. Then attempt the questions under exam conditions. When you check a solution, ask: *what was the one idea, and what in the problem should have pointed me to it?* Write that down. That sentence is what you'll remember in the exam.

The single most useful habit: before anything else, try small cases ($n = 1, 2, 3, 4$) and look for the pattern.

Questions marked with a competition (IMO 1972, IMO 1998) are past olympiad problems. The rest are classics that every olympiad student is expected to know.

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0 The toolkit: how to attack a problem

Read this first, and again at the end.

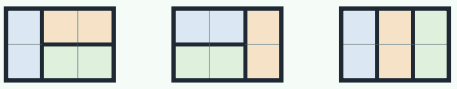
Algebra and geometry problems usually announce their method: you see an inequality, you think AM–GM; you see a cyclic quadrilateral, you chase angles. Combinatorics problems often don't. They are short, easy to understand, and the difficulty lies entirely in finding the right way to look at them. This section is about how to find that way. Come back to it after you've worked through the rest of the pack: it will make more sense then.

Step 1. Play with small cases

Before trying to prove anything, work out the answer for $n = 1, 2, 3, 4$ by hand, drawing pictures. This does three things. It makes sure you understand the question (many errors come from misreading it). It gives you a conjecture for the answer. And, most importantly, *the way you count the small cases often is the proof.* If you notice yourself reasoning "the case $n = 4$ is the case $n = 3$ plus something", that's a recursion (Section 8).

Worked example 0.1 small cases find the proof

> In how many ways can a $2 \times n$ strip be tiled with 1×2 dominoes?



The three tilings of a 2×3 strip.

Small cases. Draw them. $n = 1$: one way (a single vertical domino). $n = 2$: two ways (two vertical, or two horizontal). $n = 3$: three ways. $n = 4$: five ways. The sequence $1, 2, 3, 5$ looks like Fibonacci.

Turning the pattern into a proof. While drawing $n = 4$ you may notice that every tiling starts at the left end in one of two ways: a single vertical domino, or two horizontal dominoes stacked on top of each other (one horizontal domino on its own would leave a gap below it that only another horizontal domino can fill). In the first case what's left is a $2 \times (n-1)$ strip; in the second, a $2 \times (n-2)$ strip. So if t_n is the number of tilings,

$$t_n = t_{n-1} + t_{n-2}, \qquad t_1 = 1,\ t_2 = 2,$$

which gives $1, 2, 3, 5, 8, 13, \ldots$, the Fibonacci numbers. The small cases didn't just suggest the answer: they showed us *why* it's true.

Step 2. Recognise the type of question

The wording of the question is the best clue to the method. Learn this table: it's the map for the whole pack.

The question says…	Try first	Section
"How many…?"	bijections, complements, inclusion–exclusion, recursion, double counting	1, 2, 7, 8
"Prove that there exist…"	pigeonhole, extremal principle	3, 5
"Is it possible to reach / tile…?"	invariants, colourings	4, 6
"Show the process stops"	monovariants	5

"Who wins?"	winning/losing positions, symmetry, strategy stealing	10
"Find the maximum / minimum"	bound *and* construction	11
People, friendships, roads, matches	draw a graph	9

Step 3. Reformulate

Restate the problem in a cleaner language: a graph (people become dots, relationships become lines), a grid, a sequence of 0s and 1s, a set of subsets. A good reformulation strips away the story and often makes the problem obvious. For example, "a party where some pairs shake hands" is just a graph; "choose which of n people come" is a binary string of length n; "walk from one corner of a grid to the other" is a word made of R's and U's.

Step 4. Know what a complete answer needs

- **"Find the number"**: a number, plus an argument that every object is counted exactly once.
- **"Is it possible?"**: if yes, an explicit construction; if no, a proof (usually an invariant or colouring).
- **"Find the maximum"**: a proof that nothing larger works *and* an example achieving it. Missing either half loses most of the marks (Section 11).
- **"Who wins?"**: the winner, and their strategy, with a proof it works against *every* reply.

Step 5. Write it so a marker can follow it

Combinatorics solutions are easy to get almost right. Name things: say exactly what you are counting, what the pigeons and boxes are, what the invariant is and why each move preserves it. Short labelled paragraphs ("Bound.", "Construction.", "Claim.") make a solution easy to mark, which is good for you.

Common trap

Overcounting and undercounting are the classic errors. Whenever you count, ask: *is each object counted exactly once?* Test your method on a small case where you can list everything. When you divide by a symmetry factor ("each arrangement was counted 6 times"), check that *every* arrangement really was counted that many times.

1 Counting: bijections and stars & bars

Multiply, add, or match up with something easier.

Counting is the foundation of everything else in this pack. The goal is never to list objects one by one: it's to find a structure that lets you count them with a short calculation. Almost all counting arguments are built from three ideas: **multiply** for independent choices, **add** for separate cases, and **match up** the objects with something easier to count.

The product rule and the sum rule

If you make a sequence of choices, with a_1 options for the first, a_2 for the second (whatever you chose first), and so on, the total number of outcomes is $a_1 a_2 a_3 \cdots$. If the objects split into cases that don't overlap, the total is the sum of the cases.

Worked example 1.1 product and sum rules

> A password is 3 characters long. Each character is a lower-case letter (26 options) or a digit (10 options). (a) How many passwords are there? (b) How many contain no letter repeated and no digit repeated?

(a) Each position has 36 options, chosen independently: $36^3 = 46\,656$.

(b) Here the number of options for each position depends on what came before, so split into cases by the *pattern* of types (L = letter, D = digit). These cases don't overlap, so we add them.

- LLL: $26 \cdot 25 \cdot 24 = 15\,600$.
- Two letters, one digit: the digit's position can be chosen in 3 ways, then $26 \cdot 25 \cdot 10$, giving $3 \cdot 6500 = 19\,500$.
- One letter, two digits: $3 \cdot 26 \cdot 10 \cdot 9 = 7020$.
- DDD: $10 \cdot 9 \cdot 8 = 720$.

Total $15\,600 + 19\,500 + 7020 + 720 = 42\,840$. Notice the pattern: the **sum rule** splits into cases where the **product rule** works cleanly.

Arrangements and selections

The number of ways to arrange n different objects in a row is $n! = n(n-1)\cdots 2 \cdot 1$ (product rule: n choices for the first place, $n-1$ for the second, …). The number of ways to choose an *ordered* list of k of them is $n(n-1)\cdots(n-k+1)$.

Often we want an *unordered* selection: a set of k objects, where order doesn't matter. Each set of k objects appears $k!$ times among the ordered lists (once for each way of ordering it), so we divide:

Key fact: binomial coefficients

The number of k-element subsets of an n-element set is

$$\binom{n}{k} = \frac{n(n-1)\cdots(n-k+1)}{k!} = \frac{n!}{k!\,(n-k)!}.$$

For example, choosing 3 of 5 people: there are $5 \cdot 4 \cdot 3 = 60$ ordered lists, each group of three appears $3! = 6$ times, so $\binom{5}{3} = 10$.

Useful identities, each with a one-line reason:

- $\binom{n}{k} = \binom{n}{n-k}$: choosing the k who are in is the same as choosing the $n-k$ who are out.
- $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ (Pascal's rule): a k-subset either contains the element n (then choose $k-1$ more from the other $n-1$) or it doesn't (choose all k from the other $n-1$).
- $\sum_k \binom{n}{k} = 2^n$: both sides count all subsets of an n-set; on the right, each element is either in or out.

The bijection principle

A **bijection** is a perfect matching between two sets: every object in the first set is paired with exactly one object in the second, and vice versa. If such a matching exists, the two sets have the same size. This is the most powerful idea in counting: *to count something hard, match it with something easy.*

For example, subsets of $\{1, \ldots, n\}$ match up with binary strings of length n: write a 1 in position i if i is in the subset. The subset $\{1, 3, 4\}$ of $\{1, \ldots, 5\}$ becomes 10110. There are 2^n strings, so 2^n subsets. To prove a map is a bijection, check you can go **both ways**: every object gives a valid partner, and from the partner you can recover the object uniquely.

Worked example 1.2 a bijection that removes a restriction

> How many ways are there to choose 3 numbers from $\{1, 2, \ldots, 10\}$ so that no two of them are consecutive?

The idea. The restriction "no two consecutive" is awkward. Let's find a matching with something unrestricted. Write the chosen numbers as $a < b < c$. "Not consecutive" means $b \geq a + 2$ and $c \geq b + 2$. If we squeeze the gaps by sending

$$(a, b, c) \longmapsto (a,\ b-1,\ c-2),$$

the new numbers satisfy $a < b - 1 < c - 2$ (strictly increasing, but now they're allowed to be consecutive), and they lie in $\{1, \ldots, 8\}$.

Check it's a bijection. Going back: given any $x < y < z$ in $\{1, \ldots, 8\}$, the triple $(x, y+1, z+2)$ has gaps at least 2 and lies in $\{1, \ldots, 10\}$. The two maps undo each other. So the number we want is the number of 3-subsets of an 8-set: $\binom{8}{3} = 56$.

Sanity check on a small case. Choose 2 non-consecutive numbers from $\{1, 2, 3, 4\}$: the same argument gives $\binom{3}{2} = 3$, and indeed the options are $\{1, 3\}, \{1, 4\}, \{2, 4\}$. In general the answer is $\binom{n-k+1}{k}$.

Stars and bars

How many ways can you share out 7 identical sweets between 3 children (some children may get none)? Draw the 7 sweets as stars in a row and use 2 bars to split the row into 3 blocks: the first child gets the stars before the first bar, the second those between the bars, the third those after the second bar.

$\star\star$ | $\star\star\star\star$ | $\star$ means $(2, 4, 1)$ | $\star\star\star\star\star$ | $\star\star$ means $(0, 5, 2)$

Every row of 7 stars and 2 bars gives a sharing, and every sharing gives exactly one row. A row is 9 symbols, and it's determined by which 2 of the 9 positions hold bars: $\binom{9}{2} = 36$ ways.

Key fact: stars and bars

The number of solutions of $x_1 + x_2 + \cdots + x_k = n$ in **non-negative** integers is $\binom{n+k-1}{k-1}$ (rows of n stars and $k-1$ bars).

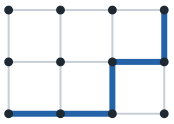
The number of solutions in **positive** integers is $\binom{n-1}{k-1}$: give each x_i one star first, then share out the remaining $n-k$ freely; equivalently, choose $k-1$ of the $n-1$ gaps between n stars in a row.

Common trap

Stars and bars is for **identical** objects going into **different** boxes. If the objects are different (7 different sweets to 3 children), each sweet independently picks a child: 3^7 ways. Always ask: are the objects identical? Are the boxes?

Lattice paths

A path on the grid from $(0, 0)$ to (m, n) using unit steps right (R) and up (U) is the same as a word with m R's and n U's. For instance, the path in the picture is RRURU. The word is fixed once you choose which m of the $m+n$ positions are R's, so there are $\binom{m+n}{m}$ paths.



The path RRURU from $(0, 0)$ to $(3, 2)$. There are $\binom{5}{2} = 10$ such paths.

Complementary counting

When the condition says "at least one", it's usually easier to count the objects with *none* and subtract from the total.

Worked example 1.3 complements beat cases

> How many 3-letter strings (using the 26 letters, repeats allowed) contain at least one A?

The tempting wrong method. "Put an A in one of 3 places, fill the other two freely: $3 \cdot 26^2 = 2028$." This overcounts: the string AAB is counted twice (once with the A chosen in position 1, once with position 2), and AAA three times.

Right method. Total strings: $26^3 = 17\,576$. Strings with no A: $25^3 = 15\,625$. So the answer is $17\,576 - 15\,625 = 1951$.

Method: which counting tool?

- Choices made one after another → **product rule**; separate cases → **sum rule**.
- Order doesn't matter → divide by the number of orderings, or use $\binom{n}{k}$ directly.
- Awkward condition → find a **bijection** to something without it.
- Identical objects into distinct boxes → **stars and bars**.
- "At least one" → **complement**.
- Proving an identity → show both sides count the same set in two ways (**committee argument**; see also Section 7).

Quick check

1. How many ways can 10 identical coins be shared among 4 people so that everyone gets at least one?
2. How many paths of R and U steps go from $(0,0)$ to $(5,3)$?
3. How many 5-digit numbers have at least one digit equal to 7?

Answers: 1. $\binom{9}{3} = 84$ · 2. $\binom{8}{3} = 56$ · 3. $90\,000 - 8 \cdot 9^4 = 37\,512$

Now practise

Each question below has its own page with space to work and a hint printed upside down at the bottom. The full solution is on the following page.

1.1 stars & bars

1.2 lattice paths

1.3 committee argument

1.4 complementary counting

Question 1.1

stars & bars

Find the number of solutions of $x_1 + x_2 + x_3 + x_4 = 20$
(a) in non-negative integers; (b) in positive integers.

Hint: (a) Stars and bars directly. (b) Substitute $y_i = x_i - 1$.

Solution 1.1

stars & bars

Find the number of solutions of $x_1 + x_2 + x_3 + x_4 = 20$
(a) in non-negative integers; (b) in positive integers.

Key idea. (a) is stars and bars with $n = 20$, $k = 4$. (b) reduces to (a) by subtracting 1 from each variable, which is a bijection between the two solution sets.

(a) Each solution corresponds to a row of 20 stars and 3 bars, with x_i the number of stars in the i-th block. Conversely, each such row gives a solution. Choosing the positions of the 3 bars among 23 positions:

$$\binom{23}{3} = \frac{23 \cdot 22 \cdot 21}{6} = 1771.$$

(b) The map $y_i = x_i - 1$ is a bijection from positive solutions of $x_1 + \cdots + x_4 = 20$ to non-negative solutions of $y_1 + \cdots + y_4 = 16$ (it can be undone by $x_i = y_i + 1$). By stars and bars there are $\binom{19}{3} = \frac{19 \cdot 18 \cdot 17}{6} = 969$ of these.

Answer: (a) 1771; (b) 969. In general, $\binom{n+k-1}{k-1}$ and $\binom{n-1}{k-1}$.

Question 1.2

lattice paths

A path from $(0, 0)$ to $(4, 4)$ consists of unit steps, each either right or up. How many such paths do *not* pass through the point $(2, 2)$?

Hint: Count all paths, then subtract those through $(2, 2)$, which split into two independent halves.

Solution 1.2

lattice paths

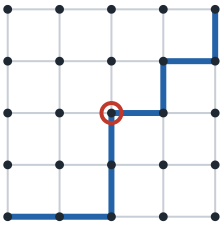
A path from $(0,0)$ to $(4,4)$ consists of unit steps, each either right or up. How many such paths do *not* pass through the point $(2,2)$?

Key idea. Complementary counting. A path through $(2,2)$ is a path to $(2,2)$ followed by a path from $(2,2)$, so those paths are counted by the product rule.

All paths. A path is a word of 4 R's and 4 U's, and every such word gives a path. So there are $\binom{8}{4} = 70$.

Paths through $(2,2)$. Such a path is a path $(0,0) \to (2,2)$ followed by a path $(2,2) \to (4,4)$, chosen independently, and every such pair gives a path through $(2,2)$. So there are $\binom{4}{2} \cdot \binom{4}{2} = 6 \cdot 6 = 36$.

Answer. $70 - 36 = 34$.



A path through $(2,2)$: the part before and the part after are chosen independently.

Answer: 34.

Question 1.3

committee argument

Prove that for every $n \geq 0$,

$$\binom{n}{0}^2 + \binom{n}{1}^2 + \cdots + \binom{n}{n}^2 = \binom{2n}{n}.$$

Hint: Count committees of size n chosen from n boys and n girls. Split by the number of boys.

Solution 1.3

committee argument

Prove that for every $n \geq 0$,

$$\binom{n}{0}^2 + \binom{n}{1}^2 + \cdots + \binom{n}{n}^2 = \binom{2n}{n}.$$

Key idea. Both sides count the n-element subsets of a set of $2n$ people. The left side sorts them by how many come from the first half.

Take n boys and n girls. The number of ways to choose n people is $\binom{2n}{n}$.

Now count by the number k of boys chosen ($0 \leq k \leq n$). Choose the k boys in $\binom{n}{k}$ ways, and the $n-k$ girls in $\binom{n}{n-k} = \binom{n}{k}$ ways. So there are $\binom{n}{k}^2$ committees with exactly k boys. Every committee has exactly one value of k, so summing over k counts each committee exactly once:

$$\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}. \qquad \blacksquare$$

Question 1.4

complementary counting

How many four-digit numbers (from 1000 to 9999) have at least one digit appearing more than once?

Hint: "At least one repeat" is the complement of "all digits different".

Solution 1.4

complementary counting

How many four-digit numbers (from 1000 to 9999) have at least one digit appearing more than once?

Key idea. Counting "at least one repeat" directly needs many cases. Its complement, "all four digits distinct", is a single product-rule count.

Total. There are $9999 - 1000 + 1 = 9000$ four-digit numbers.

All digits distinct. The first digit has 9 choices (not 0). The second has 9 (any digit except the first, now including 0), the third 8, the fourth 7. So there are $9 \cdot 9 \cdot 8 \cdot 7 = 4536$.

Answer. $9000 - 4536 = 4464$.

Answer: 4464.

2 Inclusion–exclusion

Counting objects that avoid several bad properties at once.

Complementary counting (Section 1) handles one bad property: count everything, subtract the bad. Inclusion–exclusion handles *several* bad properties at once. The difficulty is that the bad sets overlap, and the formula is the systematic way to correct for that.

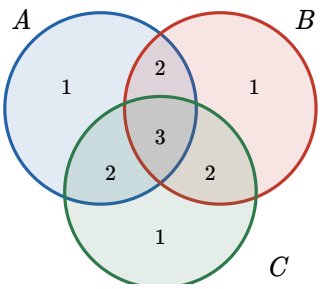
Two sets

In a class of 30, 18 play football, 15 play tennis and 8 play both. How many play at least one? Adding $18 + 15 = 33$ counts the 8 who play both *twice*, so subtract them once: $33 - 8 = 25$. Then $30 - 25 = 5$ play neither.

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Three sets

For three sets, start with $|A| + |B| + |C|$. The numbers in the picture show how many times each region has been counted so far: regions in two sets are counted twice, the centre three times.



How often each region is counted by $|A| + |B| + |C|$.

Subtracting $|A \cap B| + |B \cap C| + |C \cap A|$ fixes the double regions (now counted once) but takes the centre down from 3 to $3 - 3 = 0$. So we add $|A \cap B \cap C|$ back:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|.$$

Key fact: inclusion–exclusion

For finite sets $A_1, \dots, A_m$ inside a set of N objects,

$$|A_1 \cup \cdots \cup A_m| = \sum_i |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| - \cdots$$

and so the number of objects in **none** of the sets is

$$N - \sum_i |A_i| + \sum_{i<j} |A_i \cap A_j| - \sum_{i<j<k} |A_i \cap A_j \cap A_k| + \cdots$$

In words: add the singles, subtract the pairs, add the triples, subtract the quadruples, …

Why it works: each element is counted exactly once

Take an element lying in exactly $t \geq 1$ of the sets. It is counted $\binom{t}{1}$ times among the singles, $\binom{t}{2}$ times among the pairs, and so on, so in total

$$\binom{t}{1} - \binom{t}{2} + \binom{t}{3} - \cdots = 1 - (1-1)^t = 1$$

time, using the binomial theorem. An element in none of the sets is counted 0 times. Check with $t = 3$: $3 - 3 + 1 = 1$. ✓

Worked example 2.1 divisibility

> How many of the numbers $1, 2, \ldots, 100$ are divisible by 2 or by 3? How many by neither?

Let A be the multiples of 2 and B the multiples of 3. Then $|A| = 50$ and $|B| = \lfloor 100/3 \rfloor = 33$. A number is in both exactly when it's a multiple of 6, so $|A \cap B| = \lfloor 100/6 \rfloor = 16$. Hence $|A \cup B| = 50 + 33 - 16 = 67$, and $100 - 67 = 33$ numbers are divisible by neither.

Worked example 2.2 stars and bars with an upper limit

> How many solutions does $x_1 + x_2 + x_3 = 10$ have in integers with $0 \leq x_i \leq 4$?

Set-up. Without the upper limit, stars and bars gives $\binom{12}{2} = 66$ solutions. The bad property A_i is "$x_i \geq 5$". We want the solutions in none of A_1, A_2, A_3.

Singles. Solutions with $x_1 \geq 5$: put $x_1 = 5 + y_1$, then $y_1 + x_2 + x_3 = 5$ with all variables ≥ 0, which has $\binom{7}{2} = 21$ solutions. By symmetry $|A_i| = 21$ for each i.

Pairs. $x_1 \geq 5$ and $x_2 \geq 5$ forces $x_1 = x_2 = 5$, $x_3 = 0$: one solution. So $|A_i \cap A_j| = 1$.

Triples. All three ≥ 5 would sum to at least 15: impossible, so 0.

Answer. $66 - 3 \cdot 21 + 3 \cdot 1 - 0 = 6$.

Check by listing. With each part at most 4 and total 10, the parts are $\{4, 4, 2\}$ or $\{4, 3, 3\}$ in some order: $3 + 3 = 6$ solutions. ✓

Method: how to set it up

1. Decide what the **bad properties** are, so that the objects you want are those with *none* of them.
2. Check that the **intersections** are easy to count: "has bad properties i and j" should be a clean condition.
3. Look for **symmetry**: if any j of the m properties are satisfied by N_j objects, regardless of which j, then the answer is

$$\sum_{j=0}^{m} (-1)^j \binom{m}{j} N_j,$$

because there are $\binom{m}{j}$ ways to choose which j properties.

Typical uses: divisibility by several primes, onto functions, derangements (no object in its own place), and upper limits in stars and bars.

Common trap

Don't stop after the pairs. With three or more sets you must keep going (add triples, subtract quadruples, ...) until the intersections are empty. Also make sure an intersection is computed correctly: "multiple of 4 and of 6" means multiple of 12, not of 24 (use the lcm).

Quick check

1. How many integers from 1 to 60 are divisible by 3 or 5?
2. How many ways can 4 people swap hats so that nobody gets their own hat back?
3. How many solutions does $x_1 + x_2 + x_3 = 7$ have with $0 \le x_i \le 3$?

Answers: 1. $20 + 12 - 4 = 28$. 2. $9 \cdot 2 = 18$. 3. $\binom{9}{2} - 3\binom{5}{2} = 36 - 30 = 6$.

Now practise

Each question below has its own page with space to work and a hint printed upside down at the bottom. The full solution is on the following page.

2.1 divisibility

2.2 surjections

2.3 derangements

Question 2.1

divisibility

How many integers from 1 to 1000 are divisible by none of 2, 3, 5?

Hint: Let A_p be the multiples of p. The multiples of both 2 and 3 are the multiples of 6, and $|A_d| = \lfloor 1000/d \rfloor$.

Solution 2.1

divisibility

How many integers from 1 to 1000 are divisible by none of 2, 3, 5?

Key idea. Let the bad sets be A_2, A_3, A_5 (multiples of each). Their intersections are multiples of $6, 10, 15, 30$, which are easy to count.

Write A_d for the multiples of d in $\{1, \dots, 1000\}$, so $|A_d| = \lfloor 1000/d \rfloor$. Since $2, 3, 5$ are coprime, $A_2 \cap A_3 = A_6$, and so on. We have

$$|A_2| = 500,\ |A_3| = 333,\ |A_5| = 200, \quad |A_6| = 166,\ |A_{10}| = 100,\ |A_{15}| = 66, \quad |A_{30}| = 33.$$

By inclusion–exclusion,

$$|A_2 \cup A_3 \cup A_5| = (500 + 333 + 200) - (166 + 100 + 66) + 33 = 734.$$

So the number divisible by none of them is $1000 - 734 = 266$.

Answer: 266.

Question 2.2

How many functions from $\{1, 2, 3, 4, 5\}$ to $\{a, b, c\}$ are *onto*, i.e. take every one of the values a, b, c?

Hint: Bad property B_x: "the value x is never taken".

Solution 2.2

surjections

How many functions from $\{1, 2, 3, 4, 5\}$ to $\{a, b, c\}$ are *onto*, i.e. take every one of the values a, b, c?

Key idea. A function fails to be onto exactly when it misses some value. Counting functions that miss a given set of values is easy: they map into the remaining values.

There are $3^5 = 243$ functions in total. Let B_x be the set of functions that never take the value x.

- $|B_x| = 2^5 = 32$ (each input goes to one of the other two values).
- $|B_x \cap B_y| = 1^5 = 1$ (everything goes to the third value).
- $|B_a \cap B_b \cap B_c| = 0$.

So $|B_a \cup B_b \cup B_c| = 3 \cdot 32 - 3 \cdot 1 + 0 = 93$, and the number of onto functions is $243 - 93 = 150$.

Answer: 150. In general, the number of onto maps from an n-set to a k-set is $\sum_{j=0}^{k} (-1)^j \binom{k}{j} (k-j)^n$.

Question 2.3

derangements

n letters are put into n addressed envelopes, one in each. In how many ways can this be done so that no letter is in its correct envelope? Evaluate your formula for $n = 4$.

Hint: Bad property B_i: letter i is in envelope i. How many arrangements fix a given set of k letters?

Solution 2.3

derangements

n letters are put into n addressed envelopes, one in each. In how many ways can this be done so that no letter is in its correct envelope? Evaluate your formula for $n = 4$.

Key idea. A derangement avoids all n bad properties "letter i is correct". Fixing any particular k letters leaves $(n-k)!$ arrangements of the rest, so inclusion–exclusion gives a clean alternating sum.

Arrangements are permutations of $\{1, \dots, n\}$. Let B_i be the set of permutations with i in position i. For any k indices, the permutations fixing all of them number $(n-k)!$ (arrange the other $n-k$ letters freely), and there are $\binom{n}{k}$ ways to choose the indices. By inclusion–exclusion (the symmetric form), the number D_n of permutations in none of the B_i is

$$D_n = \sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)! = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}.$$

For $n = 4$: $D_4 = 24\left(1 - 1 + \tfrac{1}{2} - \tfrac{1}{6} + \tfrac{1}{24}\right) = 12 - 4 + 1 = 9$.

Remarks. $D_n / n! \to 1/e$, so about 37% of arrangements are derangements. The recurrence $D_n = (n-1)(D_{n-1} + D_{n-2})$ is also worth knowing; it can be proved by looking at where letter n goes.

Answer: $D_n = n!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \cdots + \frac{(-1)^n}{n!}\right)$; $D_4 = 9$.

3 The pigeonhole principle

More pigeons than boxes: proving existence without finding the object.

The pigeonhole principle sounds too obvious to be useful: if you put more pigeons than boxes, two pigeons share a box. But it is one of the most powerful tools for "prove that there exist…" problems, because it proves that something exists *without finding it*. All the skill is in choosing the boxes.

Key fact: the pigeonhole principle

If more than k objects are put into k boxes, some box contains at least two objects.

General form. If N objects go into k boxes, some box contains at least $\lceil N/k \rceil$ objects.

Proof. If every box had at most $\lceil N/k \rceil - 1$ objects, then since $\lceil N/k \rceil - 1 < N/k$, the total would be less than $k \cdot N/k = N$. Contradiction.

Warm-up. Among 13 people, two have birthdays in the same month (13 people, 12 months). Among 100 people, some month contains at least $\lceil 100/12 \rceil = 9$ birthdays. Among any 3 integers, two have the same parity (3 integers, 2 parities), so their sum is even.

Choosing the boxes

In a real problem nobody tells you what the boxes are. The trick is to work backwards: *what would two objects in the same box need to have in common for me to be finished?* Then design the boxes so that sharing a box gives exactly that.

Worked example 3.1 remainders as boxes

> Prove that among any 5 integers, there are 3 whose sum is divisible by 3.

Boxes. Sort the integers by remainder mod 3: boxes $0, 1, 2$.

Why these boxes? Three numbers from the same box have sum $\equiv 3r \equiv 0$. Three numbers from three different boxes have sum $\equiv 0 + 1 + 2 \equiv 0$. So we're done if either some box has 3 numbers, or every box is used.

Finish. Suppose no box has 3 numbers, so each box has at most 2. If only two boxes were used, there would be at most $2 + 2 = 4$ numbers, but we have 5. So all three boxes are used. Either way, we find three numbers with sum divisible by 3. ■

Is 5 needed? Yes: $0, 0, 1, 1$ (remainders) has no such triple. Always check whether your bound is sharp.

Worked example 3.2 prefix sums

> Prove that for any n integers $a_1, \ldots, a_n$, there is a block of consecutive terms $a_{i+1} + a_{i+2} + \cdots + a_j$ (with $i < j$) whose sum is divisible by n.

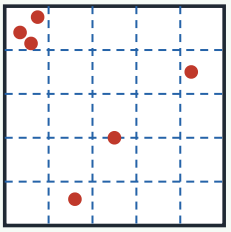
The key idea. A block sum is a difference of two prefix sums. Let $S_0 = 0$ and $S_j = a_1 + \cdots + a_j$. Then $a_{i+1} + \cdots + a_j = S_j - S_i$.

Pigeonhole. The $n + 1$ numbers $S_0, S_1, \ldots, S_n$ have only n possible remainders mod n. So two of them, S_i and S_j with $i < j$, have the same remainder. Then $n \mid S_j - S_i = a_{i+1} + \cdots + a_j$. ■

Lesson. We had n numbers but needed $n + 1$ pigeons. Including $S_0 = 0$ is the small trick that makes it work. "Differences of prefix sums" is a standard way to handle consecutive blocks.

Worked example 3.3 geometric boxes

> 51 points are placed in a unit square. Prove that some 3 of them lie inside a square of side $\frac{1}{5}$.



Cut the unit square into a 5×5 grid of 25 small squares of side $\frac{1}{5}$ (points on a dividing line can be assigned to either neighbouring square). With 51 points and 25 boxes, some small square contains at least $\lceil 51/25 \rceil = 3$ points. ■

Method: pigeonhole checklist

- **Numbers:** boxes are remainders mod k; pairs $\{1, 2\}, \{3, 4\}, \ldots$; or classes by odd part ($x = 2^a m$, m odd).
- **Geometry:** cut the region into small pieces, small enough that two points in the same piece are close.
- **Subsets:** there are 2^n subsets but only a few possible sums, so two subsets have the same sum.
- **Sequences:** label each term with a number (e.g. the length of the longest run starting there); the labels are the boxes.
- Count carefully: you need **strictly more** pigeons than boxes (or use $\lceil N/k \rceil$).
- Show the bound is **sharp**: give an example with one fewer pigeon where the conclusion fails.

Common trap

Pigeonhole tells you two objects share a box, but **not which ones**. Your argument must work for whichever pair it turns out to be. Also, the boxes must cover every possibility: in geometry, say which box a point on a boundary belongs to.

Quick check

1. How many cards must you draw from a standard 52-card deck to be sure of getting two of the same suit?
2. Show that among any 6 integers, two have a difference divisible by 5.
3. A bag has socks of 4 colours. How many must you take out to be sure of 3 of the same colour?

Answers: **1.** 5 · **2.** 6 integers, 5 remainders mod 5 · **3.** $2 + 4 = 19$.

Now practise

Each question below has its own page with space to work and a hint printed upside down at the bottom. The full solution is on the following page.

3.1 two classics

3.2 geometric pigeonhole

3.3 IMO 1972 Problem 1

3.4 Erdős–Szekeres

Question 3.1

two classics

Let S be a set of $n+1$ numbers chosen from $\{1, 2, \ldots, 2n\}$. Prove that
(a) S contains two numbers that are coprime;
(b) S contains two numbers one of which divides the other.
Show that both statements can fail for n numbers.

Hint: (a) Boxes $\{1,2\}, \{3,4\}, \ldots$. (b) Write each number as $2^k \cdot m$ with m odd; use the odd part m as the box.

Solution 3.1

two classics

Let S be a set of $n+1$ numbers chosen from $\{1, 2, \ldots, 2n\}$. Prove that
(a) S contains two numbers that are coprime;
(b) S contains two numbers one of which divides the other.
Show that both statements can fail for n numbers.

Key idea. (a) Pair up consecutive numbers, which are always coprime. (b) Sort by odd part; two numbers with the same odd part differ by a power of 2, so one divides the other.

(a) Use the n boxes $\{1,2\}, \{3,4\}, \ldots, \{2n-1, 2n\}$. With $n+1$ numbers, two are in the same box, so they are consecutive integers $t, t+1$. Any common divisor divides $(t+1) - t = 1$, so they are coprime.

(b) Write each $x \in S$ as $x = 2^k m$ with m odd. The odd part m is one of $1, 3, \ldots, 2n-1$: only n values. With $n+1$ numbers, two have the same odd part: $x = 2^k m$, $y = 2^l m$ with $k < l$. Then $x \mid y$.

Sharpness. $\{2, 4, \ldots, 2n\}$ has n elements, no two coprime. And $\{n+1, \ldots, 2n\}$ has n elements with no divisibility: if $a \mid b$ with $a < b$, then $b \geq 2a \geq 2n+2$. $\blacksquare$

Question 3.2

geometric pigeonhole

Five points are placed in a square of side 1 (possibly on the boundary). Prove that two of them are at distance at most $\frac{\sqrt{2}}{2}$.

Hint: Cut the square into four smaller squares.

Solution 3.2

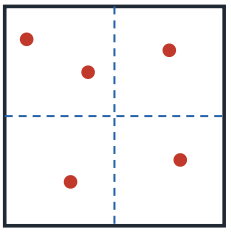
geometric pigeonhole

Five points are placed in a square of side 1 (possibly on the boundary). Prove that two of them are at distance at most $\frac{\sqrt{2}}{2}$.

Key idea. Four boxes, five points. The boxes must be small enough that two points in the same box are close: squares of side $\frac{1}{2}$ have diameter $\frac{\sqrt{2}}{2}$.

Divide the square into four closed squares of side $\frac{1}{2}$ by joining the midpoints of opposite sides. Every point lies in at least one of them; assign each point to one square containing it. Five points, four squares, so two points P, Q are in the same small square.

The greatest distance between two points of a square of side $\frac{1}{2}$ is its diagonal, $\frac{\sqrt{2}}{2}$. So $PQ \le \frac{\sqrt{2}}{2}$. ■



Sharpness. The four corners and the centre show that $\frac{\sqrt{2}}{2}$ cannot be improved.

Question 3.3

IMO 1972 Problem 1

Prove that from any set of ten distinct two-digit numbers, one can select two disjoint non-empty subsets with the same sum.

Hint: Count the subsets and the possible values of their sums.

Solution 3.3

IMO 1972 Problem 1

Prove that from any set of ten distinct two-digit numbers, one can select two disjoint non-empty subsets with the same sum.

Key idea. There are more non-empty subsets (1023) than possible sums (at most 981), so two subsets have the same sum. Removing their common elements makes them disjoint without changing the equality.

Pigeonhole. The set has $2^{10} - 1 = 1023$ non-empty subsets. Each subset sum is at least 10 and at most $10 \cdot 99 = 990$, so there are at most 981 possible sums. Hence two different non-empty subsets $X \neq Y$ have the same sum.

Make them disjoint. Let $X' = X \setminus Y$ and $Y' = Y \setminus X$. Removing the common elements from both subtracts the same amount from each sum, so X' and Y' still have equal sums, and they are disjoint. Neither is empty: if $X' = \varnothing$ then $X \subseteq Y$ with $X \neq Y$, so Y has an extra positive element and a larger sum, a contradiction. Similarly $Y' \neq \varnothing$. $\blacksquare$

Question 3.4

Erdős–Szekeres

Prove that any sequence of $mn + 1$ distinct real numbers contains an increasing subsequence of length $m + 1$ or a decreasing subsequence of length $n + 1$.

Hint: Label each term by the length of the longest increasing subsequence starting at it.

Solution 3.4

Erdős–Szekeres

Prove that any sequence of $mn+1$ distinct real numbers contains an increasing subsequence of length $m+1$ or a decreasing subsequence of length $n+1$.

Key idea. Label each term with L_i, the length of the longest increasing subsequence starting at it. If no label exceeds m, there are only m labels for $mn+1$ terms, so $n+1$ terms share a label, and terms sharing a label must decrease.

Let the sequence be $x_1, \dots, x_{mn+1}$, and let L_i be the length of the longest increasing subsequence starting at x_i. If some $L_i \geq m+1$ we are done. Otherwise every $L_i \in \{1, \dots, m\}$.

Pigeonhole. $mn+1$ terms and m labels, so some label is shared by at least $\lceil (mn+1)/m \rceil = n+1$ terms, say $x_{i_1}, \dots, x_{i_{n+1}}$ with $i_1 < \cdots < i_{n+1}$.

Same label $\Rightarrow$ decreasing. Take $i < j$ with $L_i = L_j$. If $x_i < x_j$, then x_i followed by a longest increasing subsequence from x_j is increasing of length $L_j + 1$, so $L_i \geq L_j + 1$, a contradiction. So $x_i > x_j$ (they are distinct). Hence $x_{i_1} > x_{i_2} > \cdots > x_{i_{n+1}}$ is decreasing of length $n+1$. $\blacksquare$

Sharpness. mn terms are not enough. Take n blocks, each an increasing run of m numbers, with every block entirely below the previous one, e.g. for $m=3, n=2$: $4,5,6,1,2,3$.

4 Invariants

To show a target is unreachable, find something the moves can't change.

Many problems describe a **process**: a board of numbers, a set of coins, a robot that moves, and a rule for what one move does. Then they ask whether some target can ever be reached. If the answer is "no", you can't try every sequence of moves: there are infinitely many. Instead, you find a quantity that the moves *cannot change*.

Key fact: the invariant principle

An **invariant** is a quantity that every move leaves unchanged. If an invariant has one value at the start and a different value at the target, then the target can never be reached, whatever moves are made.

Worked example 4.1 parity of a count

> Seven coins lie on a table, all heads up. A move is to turn over exactly two coins. Can you reach a position with all seven coins tails up?

What does one move do? Turning over two coins changes the number of heads by -2 (two heads become tails), $+2$ (two tails become heads) or 0 (one of each).

Invariant. In every case the number of heads changes by an even number, so its **parity** never changes.

Compare. At the start there are 7 heads (odd). The target has 0 heads (even). So the target is unreachable.

Worked example 4.2 a colouring invariant

> A knight starts on a corner of a chessboard. Can it return to the same square after exactly 25 moves?

Every knight move goes 2 squares one way and 1 the other, so $i + j$ (row plus column) changes by ± 1 or ± 3, an odd number. So each move changes the colour of the knight's square. After 25 moves (odd) the knight is on the opposite colour from where it started, so it can't be on the starting square. **No.**

Here the invariant is "colour of the square, compared with the parity of the number of moves made".

How to find an invariant

Write down exactly what *one move* does to the configuration. Then try natural quantities: the sum, the number of odd entries, the sum mod k, the sum of squares, the product. Test each against your description of a move. Often nothing is quite invariant, but a **combination** is: two quantities that change by the same amount have an invariant difference.

Worked example 4.3 combining two quantities

> The numbers $1, 2, \ldots, 20$ are written on a board. A move is to erase two numbers a, b and write $a + b - 1$. What number remains after 19 moves?

What one move does. The sum S of the numbers goes down by exactly 1, and the count c of numbers goes down by exactly 1.

Invariant. So $S - c$ never changes. At the start, $S - c = 210 - 20 = 190$.

At the end. One number z remains, so $S - c = z - 1$. Hence $z - 1 = 190$, and the final number is always **191**, whatever order the moves are made in.

Why it works: a proof by invariant is airtight

Suppose Q is invariant. After any number of moves Q still has its starting value, by induction on the number of moves (each move keeps it the same). So every reachable configuration has that value of Q. A configuration with a different value of Q is never reachable. There's no need to think about clever sequences of moves at all.

Method: find what doesn't change

- Write down precisely what one move does.
- Try: **parity** of a sum or count; a sum or difference **mod** k; a **sum of squares**; a **product**; a **colouring** count (Section 6).
- If a quantity changes in a controlled way, look for a combination (a difference, a ratio) that doesn't.
- Try your candidate on small examples before writing the proof.

Common trap

An invariant proves **impossibility**. It does not, on its own, prove that a target with the right value is reachable; that needs an explicit sequence of moves. If the question is "can we reach X?" and the invariant agrees at the start and at X, you have proved nothing yet.

Quick check

1. The numbers $1, 2, \ldots, 9$ are on a board. A move: replace two numbers by their sum. Can 50 ever appear as the last number?
2. Each cell of a 3×3 grid holds 0. A move: add 1 to two cells that share a side. Can all nine cells end up equal to 1?
3. A move replaces a pair (a, b) by $(a + 2b,\ b)$. Starting from $(1, 2)$, can you reach $(4, 2)$?

Answers: **1.** No: the sum 45 is invariant · **2.** No: each move adds 2 to the total, which stays even, but 9 is odd · **3.** No: a stays odd (it changes by $2b$)

Now practise

Each question below has its own page with space to work and a hint printed upside down at the bottom. The full solution is on the following page.

4.1 parity

4.2 mod 3

4.3 sum of squares

4.4 product invariant

Question 4.1

parity

The numbers $1, 2, \ldots, 10$ are written on a board. A move: erase two numbers a, b and write $|a - b|$. After nine moves one number remains. Prove that it is odd.

Hint: How does the parity of the sum of all numbers on the board change?

Solution 4.1

parity

The numbers $1, 2, \dots, 10$ are written on a board. A move: erase two numbers a, b and write $|a - b|$. After nine moves one number remains. Prove that it is odd.

Key idea. $a + b$ and $|a - b|$ have the same parity, so the parity of the total sum never changes.

Let S be the sum of the numbers on the board. A move replaces $a + b$ by $|a - b|$ in S. Since $a + b - |a - b| = 2\min(a, b)$ is even, the parity of S is unchanged. Initially $S = 1 + \cdots + 10 = 55$, which is odd. At the end S is the single remaining number, so it is odd. $\blacksquare$

Question 4.2

mod 3

An island has 13 red, 15 green and 17 blue chameleons. When two chameleons of different colours meet, both change to the third colour. Can all the chameleons eventually be the same colour?

Hint: Look at the differences of the counts, mod 3.

Solution 4.2

mod 3

An island has 13 red, 15 green and 17 blue chameleons. When two chameleons of different colours meet, both change to the third colour. Can all the chameleons eventually be the same colour?

Key idea. Each meeting changes the counts by $(-1, -1, +2)$ in some order. Every pairwise difference then changes by 0 or ± 3, so the differences mod 3 are invariant.

Let the counts be (r, g, b). A meeting of red and green gives $(r-1, g-1, b+2)$. Then $r - g$ is unchanged, and $g - b$ and $r - b$ change by -3. The other two kinds of meeting are similar. So every difference $r - g$, $g - b$, $r - b$ is invariant mod 3.

Initially $r - g = -2 \equiv 1$, $g - b = -2 \equiv 1$, $r - b = -4 \equiv 2 \pmod 3$. None is 0.

If all 45 chameleons were one colour, two of the counts would be 0, and their difference would be $0 \equiv 0 \pmod 3$. Contradiction.

Answer: No.

Question 4.3

sum of squares

A pair of real numbers (a, b) may be replaced by $\left(\dfrac{a+b}{\sqrt{2}},\ \dfrac{a-b}{\sqrt{2}}\right)$. Starting from $(3, 4)$, can we reach $(4, 6)$?

Hint: Compute $a^2 + b^2$ before and after a move.

Solution 4.3

sum of squares

A pair of real numbers (a, b) may be replaced by $\left(\dfrac{a+b}{\sqrt{2}},\ \dfrac{a-b}{\sqrt{2}}\right)$. Starting from $(3, 4)$, can we reach $(4, 6)$?

Key idea. The move is a rotation followed by a reflection, which preserves distance from the origin, so $a^2 + b^2$ is invariant.

$$\left(\frac{a+b}{\sqrt{2}}\right)^2 + \left(\frac{a-b}{\sqrt{2}}\right)^2 = \frac{(a^2 + 2ab + b^2) + (a^2 - 2ab + b^2)}{2} = a^2 + b^2.$$

So $a^2 + b^2$ is invariant. It equals $9 + 16 = 25$ at the start and $16 + 36 = 52$ at the target.

Answer: No.

Question 4.4

product invariant

The numbers $1, 2, \ldots, n$ are written on a board. A move: erase two numbers a, b and write $a + b + ab$. Find the number that remains at the end, and show it doesn't depend on the order of the moves.

Hint: $a + b + ab = (1 + a)(1 + b) - 1$.

Solution 4.4

product invariant

The numbers $1, 2, \dots, n$ are written on a board. A move: erase two numbers a, b and write $a + b + ab$. Find the number that remains at the end, and show it doesn't depend on the order of the moves.

Key idea. $a + b + ab + 1 = (1 + a)(1 + b)$. So the product of $(1 + x)$ over all numbers x on the board is unchanged by every move.

Let $P = \prod(1 + x)$, over all numbers x on the board. A move removes the factors $(1 + a)(1 + b)$ and adds the factor $1 + (a + b + ab) = (1 + a)(1 + b)$. So P is invariant. Initially $P = 2 \cdot 3 \cdots (n + 1) = (n + 1)!$. At the end, with one number z left, $P = 1 + z$. So $z = (n + 1)! - 1$, whatever the order of moves.

Answer: $(n + 1)! - 1$.

5 The extremal principle and monovariants

Extreme objects give free information; one-way quantities force processes to stop.

Two closely related ideas. The **extremal principle**: look at the biggest, smallest, longest or "best" object, and use the fact that it is extreme. The **monovariant**: to show a process stops, find a quantity that always moves in one direction and can't do so for ever.

The extremal principle

If you are asked to prove that something exists, or that a configuration has some property, consider an *extreme* one: the player with the most wins, the longest path, the arrangement with the smallest total. Being extreme gives you extra information for free, and it's often exactly the information you need.

Worked example 5.1 the player with the most wins

> In a tournament every pair of players plays once, and there are no draws. Prove that there is a player X such that every other player either lost to X, or lost to someone who lost to X.

Pick the extreme object. Let X be a player with the most wins (there are finitely many players, so this exists).

Use it. Suppose some player Y did *not* lose to X, and did not lose to anyone who lost to X. Then Y beat X, and Y also beat every player that X beat (since Y didn't lose to any of them). So Y has at least all of X's wins, plus the win against X: more wins than X. That contradicts the choice of X. ■

Lesson. We never had to *find* X by searching. Choosing the extreme object and arguing by contradiction did all the work.

Worked example 5.2 a longest path

> In a finite graph, every vertex has degree at least 2. Prove that the graph contains a cycle.

Take a **longest path** $v_0 v_1 \dots v_\ell$ (paths don't repeat vertices; there are finitely many, so a longest one exists). The end v_0 has at least 2 neighbours, so it has a neighbour $w \neq v_1$. If w were not on the path, $w v_0 v_1 \dots v_\ell$ would be a longer path. So $w = v_i$ for some $i \geq 2$, and $v_0 v_1 \dots v_i v_0$ is a cycle. ■

Common trap

Make sure the extreme object **exists**. Any finite set has a largest and smallest element, and any non-empty set of positive integers has a smallest element. But an infinite set of real numbers may have none: $\{1, \tfrac{1}{2}, \tfrac{1}{3}, \dots\}$ has no smallest element. Say in your write-up why the extreme object exists.

Monovariants

A **monovariant** is a quantity that changes in only one direction: every move strictly increases it (or strictly decreases it). If, in addition, it can take only finitely many values (or it's a non-negative integer that strictly decreases), then the process **must stop**.

Worked example 5.3 swapping out-of-order neighbours

> A row of n different numbers is given. A move: find two *adjacent* numbers in the wrong order (bigger on the left) and swap them. Prove that whatever moves are chosen, the process stops, and that it ends with the numbers in increasing order.

Monovariant. Let I be the number of **inversions**: pairs (not necessarily adjacent) in which the larger number is to the left of the smaller. Swapping an adjacent out-of-order pair fixes that pair and doesn't change the relative order of any other pair. So every move decreases I by exactly 1.

It stops. I is a non-negative integer, at most $\binom{n}{2}$ at the start, so there can be at most $\binom{n}{2}$ moves.

Where it stops. The process stops only when no adjacent pair is out of order, which means the row is increasing. ■

Why it works: monovariants force termination

A strictly decreasing sequence of non-negative integers $Q_0 > Q_1 > Q_2 > \cdots$ has at most $Q_0 + 1$ terms. More generally, a quantity that strictly increases at each step can't repeat a value, so if it can take only finitely many values, there are only finitely many steps.

The two ideas together

Very often a single proof can be written either way. "Take the configuration that minimises Q. If it weren't good, a move would decrease Q, contradiction" is the extremal form. "Keep making moves that decrease Q; this stops, and when it stops the configuration is good" is the monovariant form. Use whichever reads more cleanly.

Method: extremal and monovariant

- To prove existence: consider the **largest/smallest/longest** object, or the configuration **minimising** a cleverly chosen total, and show that if it failed you could make something even more extreme.
- To prove a process ends: find a quantity that changes **strictly in one direction** and has **finitely many values** (or is a non-negative integer).
- Good candidates: a total sum; a number of "bad pairs"; a total length; the number of inversions.

Quick check

1. Why does every finite set of points in the plane have two points at maximal distance?
2. Each of n coins is heads or tails. A move: pick a head that has a tail immediately to its right, and turn both over. Why must this stop?
3. Prove that $\sqrt{2}$ is irrational using the extremal principle.

Answers: **1.** finitely many pairs, so one has the largest distance · **2.** read H as 1, T as 0; each move replaces 10 by 01, and the binary number decreases · **3.** if $d/b = q$ with q minimal, then $b(d(/)p(-)2q = $ with $b < 0 - p < q$

Now practise

Each question below has its own page with space to work and a hint printed upside down at the bottom. The full solution is on the following page.

5.1 extremal

5.2 monovariant

5.3 extremal configuration

5.4 minimal total length

Question 5.1

extremal

Every point of the integer grid $\mathbb{Z}^2$ is labelled with a positive integer. Each label equals the average of the labels of its four neighbours. Prove that all labels are equal.

Hint: Positive integers have a minimum. What happens at a point where the minimum is attained?

Solution 5.1

extremal

Every point of the integer grid $\mathbb{Z}^2$ is labelled with a positive integer. Each label equals the average of the labels of its four neighbours. Prove that all labels are equal.

Key idea. Positive integers have a smallest value m. At a point labelled m, the four neighbours average to m but are each at least m, so all four equal m. This spreads to every point.

The labels are positive integers, so there is a smallest label m, attained at some point P. (This is where integrality matters: for positive reals the minimum might not exist.)

At any point X with label m, the four neighbours have labels $\geq m$ and average exactly m. If any neighbour were $> m$, the average would exceed m. So all four neighbours have label m.

Starting from P and moving one step at a time, every lattice point can be reached, so by induction on the number of steps, every point has label m. $\blacksquare$

Question 5.2

monovariant

Real numbers are written in an $m \times n$ table. A move: change the sign of every entry in one row, or in one column. Prove that by a sequence of moves you can make every row sum and every column sum non-negative.

Hint: Do the obvious greedy move. Show the total sum of the table strictly increases.

Solution 5.2

monovariant

Real numbers are written in an $m \times n$ table. A move: change the sign of every entry in one row, or in one column. Prove that by a sequence of moves you can make every row sum and every column sum non-negative.

Key idea. Flip any line with a negative sum. That increases the total sum of the table strictly, and the total can take only finitely many values, so the process stops, at a table with no negative line.

Procedure. While some row or column has negative sum $-s < 0$, flip it.

Monovariant. Flipping a line with sum $-s$ changes that line's sum to $+s$ and leaves all other entries alone. So the total sum T of the table increases by $2s > 0$.

Termination. Each entry is always $\pm$ its original value, so there are at most 2^{mn} possible tables and T takes finitely many values. A strictly increasing sequence of values from a finite set must stop, so after finitely many moves no line has negative sum. ■

Question 5.3

extremal configuration

In a parliament, each member has at most three enemies (enmity is mutual). Prove that the parliament can be split into two houses so that every member has at most one enemy in their own house.

Hint: Among all splits, take one with the fewest pairs of enemies in the same house.

Solution 5.3

extremal configuration

In a parliament, each member has at most three enemies (enmity is mutual). Prove that the parliament can be split into two houses so that every member has at most one enemy in their own house.

Key idea. Take a split minimising the number N of "enemy pairs in the same house". If someone had ≥ 2 enemies in their house, moving them across would reduce N.

There are finitely many splits, so choose one minimising N, the number of pairs of enemies sitting in the same house.

Suppose some member X has at least 2 enemies in their own house. X has at most 3 enemies altogether, so at most 1 in the other house. Move X to the other house. The enemy pairs involving X inside a house go from at least 2 to at most 1, and no other pair is affected. So N strictly decreases, contradicting minimality.

Hence in the minimising split everyone has at most one enemy in their own house. ■

Question 5.4

minimal total length

n red and n blue points are given in the plane, no three collinear. Prove that they can be joined in n red–blue pairs by segments no two of which cross.

Hint: Among all pairings, take the one with minimal total length. What would happen if two segments crossed?

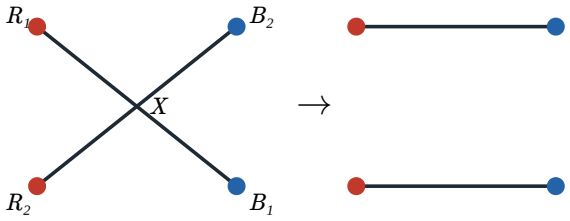
Solution 5.4

minimal total length

n red and n blue points are given in the plane, no three collinear. Prove that they can be joined in n red–blue pairs by segments no two of which cross.

Key idea. The pairing with smallest total length has no crossings. If R_1B_1 and R_2B_2 crossed, swapping partners would be strictly shorter, by the triangle inequality.

There are $n!$ red–blue pairings, so one has minimal total length. Suppose in it the segments R_1B_1 and R_2B_2 cross at a point X. Replace them by R_1B_2 and R_2B_1.



By the triangle inequality,

$$R_1B_2 + R_2B_1 < (R_1X + XB_2) + (R_2X + XB_1) = R_1B_1 + R_2B_2.$$

The inequalities are strict because R_1, X, B_2 are not collinear. If they were, B_2 would lie on line R_1X, which is line R_1B_1, putting three points on a line. Similarly for R_2, X, B_1. So the new pairing is shorter, a contradiction. Hence the minimal pairing has no crossings. ■

6 Colouring and tiling

Invariants for tiling problems.

Colouring is an invariant argument for **tiling and covering** problems. You colour the board in a clever way so that every tile, wherever it is placed, covers the same number of squares of each colour. Then you compare with the colour counts of the region you're trying to cover. If they don't match, no tiling exists.

The chessboard colouring

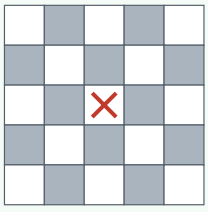
Colour the squares black and white like a chessboard: square (i, j) (row i, column j) is black when $i + j$ is even. Two squares that share a side have different colours. So a **domino** (1×2 or 2×1), wherever it is placed, covers exactly one black and one white square. Therefore:

Key fact: the domino count

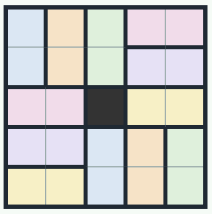
If a region can be tiled by dominoes, it has **equally many** black and white squares. Equivalently: if the numbers are different, the region can't be tiled.

Worked example 6.1 two boards with a square removed

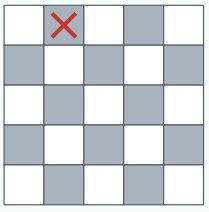
(a) Can a 5×5 board with its *centre* square removed be tiled by dominoes?
(b) What if instead we remove the square in row 1, column 2?



(a) centre removed



a tiling of (a)



(b) a white square removed

Count colours. A 5×5 board has 13 squares with $i + j$ even (black, including all four corners) and 12 white.

(b) first. The square $(1, 2)$ has $i + j = 3$, odd, so it's white. Removing it leaves 13 black and 11 white. Each domino covers one of each, so 12 dominoes would cover 12 of each. Impossible: **no**.

(a) The centre $(3, 3)$ is black. Removing it leaves 12 black and 12 white. The colouring does *not* rule out a tiling here, but it doesn't prove one exists either. For that we need to **construct** one, and the picture shows one: four 2×3 rectangles arranged like a pinwheel, each made of 3 dominoes. So the answer is **yes**.

Common trap

A colouring argument proves **impossibility** only. Matching colour counts never prove that a tiling exists: for that you must give a tiling. (For example, a 2×2 board with two diagonally opposite squares removed has one black and one white square left, but can't be covered by a domino.)

Colourings for longer tiles

For a $1 \times k$ strip, use k colours: colour square (i, j) with $(i + j) \bmod k$. Along any row or column, the colours cycle $0, 1, \ldots, k-1, 0, 1, \ldots$, so any k consecutive squares in a line contain **one of each colour**. Wherever a $1 \times k$ strip is placed, it covers one square of each colour.

0	1	2	0	1	2	0
1	2	0	1	2	0	1
2	0	1	2	0	1	2
0	1	2	0	1	2	0

The colouring $(i + j) \bmod 3$: every 1×3 strip, horizontal or vertical, covers one 0, one 1 and one 2.

Worked example 6.2 an unbalanced tile

> Can a 6×6 board be tiled by T-tetrominoes (four squares in a T shape)?

Count. The area is 36, so a tiling would use 9 T-tetrominoes. The board has 18 black and 18 white squares.

What one tile covers. With the chessboard colouring, a T-tetromino covers its centre square and the three squares next to it. The three neighbours all have the opposite colour to the centre. So each T covers either 3 black + 1 white, or 1 black + 3 white: it changes the difference "black minus white" by $+2$ or -2.

Parity. With 9 tiles, black minus white is a sum of nine terms each ± 2. If p tiles are $+2$, the sum is $2p - 2(9 - p) = 4p - 18$, which is never 0 because p would have to be 4.5. But the board has black minus white $= 0$. So: **no**.

Lesson. Even when tiles don't cover colours equally, the colouring still controls what's possible.

Method: colour so each tile looks the same

- **Dominoes:** chessboard colouring.
- $1 \times k$ **strips:** colour by $(i + j) \bmod k$, or by $i \bmod k$, so each strip covers one of each colour.
- 1×4 **strips:** also try colouring 2×2 *blocks* like a chessboard.
- Count each colour on the board and compare with what the tiles can cover.
- If two different colourings each give a condition, **combine** them.
- To prove a tiling **exists**, draw one, often by building it from smaller tileable pieces or by induction.

Quick check

1. Can a 3×3 board with one corner removed be tiled by dominoes?
2. Can a 7×7 board with its centre square removed be tiled by 1×2 dominoes?
3. How many black squares does a 7×7 chessboard-coloured board have, if the corners are black?

Answers: **1.** Yes: one domino for the rest of row 1, one for the rest of column 1, two for the $22 \times$ block left · **2.** Yes: colours balance (centre is black) and a pinwheel of four $34 \times$ rectangles works · **3.** 25

Now practise

Each question below has its own page with space to work and a hint printed upside down at the bottom. The full solution is on the following page.

6.1 dominoes

6.2 1×4 strips

6.3 two colourings

Question 6.1

dominoes

Two diagonally opposite corner squares are removed from an 8×8 chessboard. Can the remaining 62 squares be covered by 31 dominoes (1×2 rectangles)?

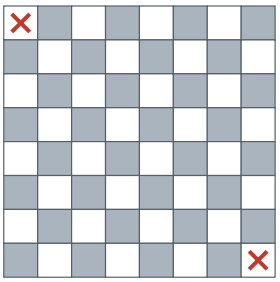
Hint: What colour are the removed squares?

Solution 6.1

dominoes

Two diagonally opposite corner squares are removed from an 8×8 chessboard. Can the remaining 62 squares be covered by 31 dominoes (1×2 rectangles)?

Key idea. Every domino covers one black and one white square. Opposite corners have the same colour, so the colours are unbalanced.



Colour the board as a chessboard. Opposite corners have the same colour (say white), since moving from one to the other changes the row by 7 and the column by 7, so $i + j$ changes by 14. The remaining board has 32 black and 30 white squares. Every domino covers two adjacent squares, one black and one white, so 31 dominoes cover exactly 31 of each. Since $32 \neq 31$, this is impossible.

Answer: No.

Question 6.2

1×4 strips

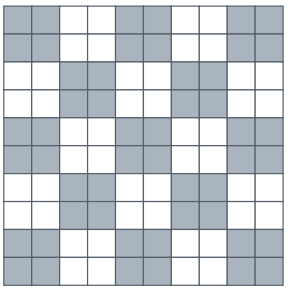
Can a 10×10 board be tiled with 1×4 rectangles?

Hint: Divide the board into 2×2 blocks and colour the blocks like a chessboard.

Solution 6.2

Can a 10×10 board be tiled with 1×4 rectangles?

Key idea. With 2×2 blocks coloured alternately, every 1×4 strip covers exactly 2 black and 2 white cells. The 10×10 board has 5×5 blocks, an odd number, so the colours are unbalanced.



Number rows and columns 0 to 9, and colour cell (i, j) black if $\lfloor i/2 \rfloor + \lfloor j/2 \rfloor$ is even. This colours the 2×2 blocks like a chessboard.

Each strip covers 2 + 2. A horizontal strip covers 4 consecutive columns $j, \ldots, j+3$ in one row. The values $\lfloor j/2 \rfloor$ along it are two consecutive integers, each appearing twice, so it covers 2 black and 2 white cells. Vertical strips are the same.

Count. There are $5 \times 5 = 25$ blocks: 13 black and 12 white. So the board has 52 black and 48 white cells. A tiling by 25 strips would cover 50 of each, so it is impossible.

Answer: No.

Question 6.3

two colourings

An 8×8 board is covered by 21 straight trominoes (1×3 rectangles) and one single square. Which squares can the single square occupy? (Number rows and columns 1 to 8.)

Hint: Colour cell (i, j) with $(i + j) \bmod 3$. Then use a second colouring, $(i - j) \bmod 3$.

Solution 6.3

two colourings

An 8×8 board is covered by 21 straight trominoes (1×3 rectangles) and one single square. Which squares can the single square occupy? (Number rows and columns 1 to 8.)

Key idea. Under the colouring $(i + j) \bmod 3$, each tromino covers one cell of each colour. The board has 22 cells of colour 0 and 21 of each other, so the single square has colour 0. The colouring $(i - j) \bmod 3$ gives the same condition. Together these force $i, j \in \{3, 6\}$.

Colouring 1. Colour (i, j) with $(i + j) \bmod 3$. A tromino covers three consecutive cells in a row or column, so $i + j$ takes three consecutive values, one in each class mod 3. Hence the 21 trominoes cover exactly 21 cells of each colour. Counting the board directly gives 22 cells of colour 0 and 21 each of colours 1 and 2. So the single square has $i + j \equiv 0 \pmod 3$.

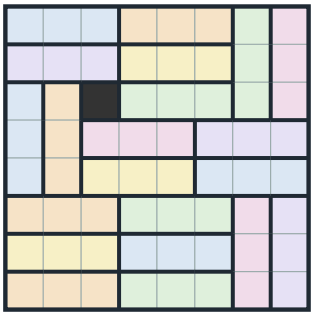
Colouring 2. By the same argument with $(i - j) \bmod 3$, which also has counts 22, 21, 21 with colour 0 the large one, the single square has $i - j \equiv 0 \pmod 3$.

Combine. Adding gives $2i \equiv 0$, so $i \equiv 0 \pmod 3$, and then $j \equiv 0 \pmod 3$. So $i, j \in \{3, 6\}$: the square is one of $(3, 3), (3, 6), (6, 3), (6, 6)$.

All four are possible. Here is a tiling with the single square at $(3, 3)$. The other three positions follow by reflecting the board.

	1	2	3	4	5	6	7	8
1	2	0	1	2	0	1	2	0
2	0	1	2	0	1	2	0	1
3	1	2	0	1	2	0	1	2
4	2	0	1	2	0	1	2	0
5	0	1	2	0	1	2	0	1
6	1	2	0	1	2	0	1	2
7	2	0	1	2	0	1	2	0
8	0	1	2	0	1	2	0	1

$(i + j) \bmod 3$



single square at $(3, 3)$

Answer: Exactly the four squares $(3, 3)$, $(3, 6)$, $(6, 3)$, $(6, 6)$.

7 Double counting

Count the same thing twice and compare.

Double counting means counting *one* collection of objects in *two* different ways. Both counts must give the same number, and setting them equal gives an equation you couldn't easily have found otherwise. It proves identities, gives answers, and, when one of the counts is only an estimate, proves inequalities.

The idea in a table

Suppose some students belong to some clubs. Make a table with a row for each student, a column for each club, and a tick when that student is in that club. The total number of ticks can be found by adding up the **row totals** (each student's number of clubs) or the **column totals** (each club's size). The two answers are equal.

	club 1	club 2	club 3	club 4	row total
Ann	✓	✓		✓	**3**
Ben		✓	✓		**2**
Cat	✓		✓	✓	**3**
column total	**2**	**2**	**2**	**2**	**8**

Here $3+2+3 = 2+2+2+2 = 8$. That's the whole principle. The art is choosing what the ticks are: usually **pairs** such as (person, club they're in), (vertex, edge at it), (pair of judges, contestant they agree on).

Worked example 7.1 finding an unknown

> Each of 20 students belongs to exactly 3 clubs, and each club has exactly 12 members. How many clubs are there?

Count the pairs (student, club) with the student in the club.

By students: $20 \times 3 = 60$. **By clubs:** if there are c clubs, $12c$.

So $12c = 60$ and $c = 5$.

Key fact: the handshake lemma

In any graph (Section 9), the sum of the degrees equals twice the number of edges:

$$\sum_{v} \deg v = 2|E|.$$

Proof. Count pairs (vertex, edge at that vertex). By vertices: $\sum \deg v$. By edges: each edge has 2 ends, so $2|E|$.

Consequence. The sum of the degrees is even. For instance, 9 people can't each know exactly 3 of the others: the degrees would sum to 27, which is odd.

Proving identities by double counting

To prove $A = B$, find a set and show that A counts it one way and B counts it another way. This is often called a **combinatorial proof**, or a committee argument.

Worked example 7.2 pairs sorted by their larger element

> Prove that $1 + 2 + \cdots + (n-1) = \binom{n}{2}$.

Count the 2-element subsets of $\{1, \ldots, n\}$. Directly, there are $\binom{n}{2}$.

Now sort them by their **larger element** k. If the larger element is k, the smaller can be any of $1, \ldots, k-1$: $k-1$ choices. So the number of pairs is $\sum_{k=2}^{n}(k-1) = 1 + 2 + \cdots + (n-1)$. ■

Pattern. One side counts the set directly. The other side splits it into cases (here by the largest element) and adds them up. That's the most common shape of a combinatorial proof.

Inequalities by double counting

If one count is exact and the other is only **bounded**, you get an inequality instead of an equation.

Worked example 7.3 a bound from counting pairs

> Ten students form committees. Each committee has exactly 4 members, and no two students serve together on more than one committee. Prove there are at most 7 committees.

Count the pairs (pair of students, committee containing both).

By committees: each committee contains $\binom{4}{2} = 6$ pairs of students, so the count is exactly $6c$.

By pairs of students: there are $\binom{10}{2} = 45$ pairs, and each is together in at most one committee, so the count is at most 45.

So $6c \le 45$, giving $c \le 7$. ■

(This only proves a bound. Whether 7 is actually achievable is a separate question; see Section 11.)

Method: count the same thing two ways

- Choose objects to count: usually pairs or triples such as (person, committee), (vertex, edge), (pair, set containing it).
- Count them by summing over the first coordinate, then over the second.
- **Identities:** both counts are exact. **Inequalities:** one count is exact and the other is bounded, often by the condition in the problem, or by an inequality such as AM–GM or Cauchy–Schwarz.
- A useful fact for the bounding step: $\binom{x}{2} + \binom{b-x}{2}$ is smallest when x is as close to $b/2$ as possible.

Quick check

1. A graph has 10 vertices, each of degree 4. How many edges does it have?
2. Prove $\binom{n}{k} \cdot k = n\binom{n-1}{k-1}$ by counting committees with a chair.
3. In a group, every person has exactly 3 friends and there are 12 friendships. How many people are there?

Answers: **1.** 20 · **2.** choose the committee then the chair, or the chair then the rest · **3.** 8

Now practise

Each question below has its own page with space to work and a hint printed upside down at the bottom. The full solution is on the following page.

7.1 handshakes

7.2 committee with a chair

7.3 IMO 1998 Problem 2

Question 7.1

handshakes

At a party, some pairs of people shake hands. Prove that the number of people who shake an odd number of hands is even.

Hint: Add up the numbers of handshakes of all people. How many times is each handshake counted?

Solution 7.1

handshakes

At a party, some pairs of people shake hands. Prove that the number of people who shake an odd number of hands is even.

Key idea. Summing each person's handshake count counts every handshake twice, so the total is even. An even total can't have an odd number of odd terms.

Let person i shake d_i hands and let E be the number of handshakes. Count pairs (person, handshake they took part in). Summing over people gives $\sum d_i$. Summing over handshakes gives $2E$, since each handshake involves two people. So $\sum d_i = 2E$ is even.

Split the sum into even d_i and odd d_i. The even ones sum to an even number, so the odd ones must too. A sum of odd numbers is even only if there is an even number of them. ■

Question 7.2

committee with a chair

Prove that

$$\sum_{k=0}^{n} k\binom{n}{k} = n\,2^{n-1}.$$

Hint: Count committees (of any size) with a chair, chosen from n people.

Solution 7.2

committee with a chair

Prove that

$$\sum_{k=0}^{n} k\binom{n}{k} = n\,2^{n-1}.$$

Key idea. Count pairs (committee, chair of that committee). Choosing the committee first gives the left side; choosing the chair first gives the right side.

Count the pairs (C, p) where C is a subset of the n people and $p \in C$ is its chair.

Committee first. For each k there are $\binom{n}{k}$ committees of size k, each with k choices of chair. Total $\sum_k k\binom{n}{k}$.

Chair first. Choose the chair in n ways, then any subset of the other $n-1$ people to complete the committee: 2^{n-1} ways. Total $n2^{n-1}$.

The two counts are equal. ■

Question 7.3

IMO 1998 Problem 2

In a competition there are a contestants and b judges, where $b \geq 3$ is odd. Each judge rates each contestant as "pass" or "fail". Suppose k is a number such that any two judges agree on at most k contestants. Prove that

$$\frac{k}{a} \geq \frac{b-1}{2b}.$$

Hint: Count triples (contestant, unordered pair of judges who agree on that contestant). Bound the count from above using k, and from below by looking at each contestant.

Solution 7.3

IMO 1998 Problem 2

In a competition there are a contestants and b judges, where $b \geq 3$ is odd. Each judge rates each contestant as "pass" or "fail". Suppose k is a number such that any two judges agree on at most k contestants. Prove that

$$\frac{k}{a} \geq \frac{b-1}{2b}.$$

Key idea. Count N, the number of (pair of judges, contestant) with the pair agreeing. Per pair of judges, $N \leq k\binom{b}{2}$. Per contestant, if x judges pass them there are $\binom{x}{2} + \binom{b-x}{2} \geq \frac{(b-1)^2}{4}$ agreeing pairs. Compare.

Let N be the number of triples $(\{J, J'\}, C)$ with $J \neq J'$ judges who give contestant C the same rating.

Upper bound. Each of the $\binom{b}{2}$ pairs of judges agrees on at most k contestants, so $N \leq k\binom{b}{2} = \dfrac{kb(b-1)}{2}$.

Lower bound. Fix a contestant, passed by x judges and failed by $b - x$. The agreeing pairs number

$$\binom{x}{2} + \binom{b-x}{2} = x^2 - bx + \frac{b^2 - b}{2} = \left(x - \frac{b}{2}\right)^2 + \frac{b^2 - 2b}{4}.$$

Since b is odd and x is an integer, $\left|x - \frac{b}{2}\right| \geq \frac{1}{2}$, so this is at least $\frac{1}{4} + \frac{b^2-2b}{4} = \frac{(b-1)^2}{4}$. Summing over contestants: $N \geq \dfrac{a(b-1)^2}{4}$.

Combine. $\dfrac{kb(b-1)}{2} \geq \dfrac{a(b-1)^2}{4}$. Dividing by $\dfrac{ab(b-1)}{2} > 0$: $\dfrac{k}{a} \geq \dfrac{b-1}{2b}$. $\blacksquare$

8 Recursion and induction

Solve size n using the answers for smaller sizes.

Many counting problems are hard to solve directly for size n, but easy to solve if you already know the answers for smaller sizes. That's **recursion**. Its partner for proofs is **induction**: prove the result for small cases, then show how each case follows from smaller ones.

Recursion: look at the last step

To find a recurrence, classify the objects of size n by what happens at one end: is the element n used? What does the last column look like? What was the last move? Each class is usually a smaller copy of the original problem.

Worked example 8.1 the Tower of Hanoi

> There are 3 pegs, and n discs of different sizes on the first peg, largest at the bottom. A move takes the top disc of one peg and puts it on another peg, never on a smaller disc. What is the smallest number of moves needed to move all n discs to the third peg?

Small cases. $h_1 = 1$, $h_2 = 3$ (small disc aside, big disc across, small disc on top), $h_3 = 7$.

Look at the largest disc. At the moment the largest disc moves, the other $n-1$ discs must all be on the remaining peg (they can't be on top of it, nor on the peg it's moving to). Getting them there takes at least h_{n-1} moves. After the largest disc's move, they must all come back on top of it: at least h_{n-1} more. So $h_n \geq 2h_{n-1} + 1$, and this is achievable by doing exactly that. So $h_n = 2h_{n-1} + 1$.

Solve it. The values $1, 3, 7, 15, \ldots$ suggest $h_n = 2^n - 1$. Check by induction: $h_1 = 1 = 2^1 - 1$, and if $h_{n-1} = 2^{n-1} - 1$ then $h_n = 2(2^{n-1} - 1) + 1 = 2^n - 1$. ✓

Tip. A neat trick for this recurrence: $h_n + 1 = 2(h_{n-1} + 1)$, so $h_n + 1$ doubles each time.

Induction

To prove a statement $P(n)$ for all $n \geq 1$:

1. **Base case:** prove $P(1)$.
2. **Inductive step:** assume $P(n-1)$ (the *inductive hypothesis*) and use it to prove $P(n)$.

Then $P(1)$ holds, so $P(2)$ does, so $P(3)$ does, and so on. In combinatorics, the inductive step usually means: *take an object of size n, find a smaller object inside it, and use the hypothesis on that.* Or, for constructions: *build the size-n object out of smaller ones.*

Key fact: strong induction

Sometimes $P(n)$ follows not from $P(n-1)$ but from some earlier case. **Strong induction** lets you assume $P(k)$ for *all* $k < n$. It's just as valid: if the statement ever failed, there'd be a smallest n where it fails, and all smaller cases would be true, which the step says is impossible.

Worked example 8.2 stamps

> You have unlimited 4p and 5p stamps. Prove that every amount of $n \geq 12$ pence can be made exactly.

Base cases. $12 = 4+4+4$, $13 = 4+4+5$, $14 = 4+5+5$, $15 = 5+5+5$.

Step. Let $n \geq 16$ and assume every amount from 12 to $n-1$ can be made. Then $n-4 \geq 12$ can be made; add one 4p stamp. ■

Why four base cases? The step reaches back 4, so it only works once $n - 4 \geq 12$. The amounts $12, 13, 14, 15$ have to be checked by hand. (And 11 really can't be made, so "12" is the right starting point.)

Common trap

Check the base cases cover everything the step needs. If the step uses $P(n-1)$ and $P(n-2)$, you need **two** base cases. Also make sure the step works for the *smallest* n you apply it to: a famous fake proof ("all horses are the same colour") fails exactly because its step doesn't work from $n = 1$ to $n = 2$.

Method: build the answer from smaller cases

- **Recursion:** classify the size-n objects by what happens at the last element, and count each class using smaller sizes. Then solve the recurrence or recognise it (F_n, 2^n, $2^n - 1$, …), and prove your formula by induction.
- **Induction for constructions:** split a size-n object into smaller copies, and deal with the leftover piece.
- **Strong induction:** assume the result for all smaller cases, not just $n-1$.
- Fibonacci: $F_1 = F_2 = 1$, $F_{k+1} = F_k + F_{k-1}$, giving $1, 1, 2, 3, 5, 8, 13, 21, \ldots$

Quick check

1. How many binary strings of length n have no two consecutive 0s? Find a recurrence and the answer for $n = 5$.
2. n points on a circle are joined by all chords. Into how many pieces is the circle's interior cut when $n = 1, 2, 3, 4, 5$ (no three chords meeting at a point)? Don't trust the pattern!
3. Prove by induction that $1 + 3 + 5 + \cdots + (2n-1) = n^2$.

Answers: 1. $s \quad s + s = s$, 23; $s = s = 13$; $s = s \cdot$ **2.** $1, 2, 4, 8, 16$, but $9n =$ gives 31, not 32 · **3.** step: $n(+ (1 - n2(=)1u$

Now practise

Each question below has its own page with space to work and a hint printed upside down at the bottom. The full solution is on the following page.

8.1 Fibonacci

8.2 regions

8.3 inductive construction

8.4 strong induction

Question 8.1

Let a_n be the number of subsets of $\{1, 2, \ldots, n\}$ (including the empty set) that contain no two consecutive integers. Prove that $a_n = F_{n+2}$.

Hint: Split according to whether n is in the subset.

Solution 8.1

Let a_n be the number of subsets of $\{1, 2, \ldots, n\}$ (including the empty set) that contain no two consecutive integers. Prove that $a_n = F_{n+2}$.

Key idea. Either n is not in the subset, leaving any valid subset of $\{1, \ldots, n-1\}$; or n is in it, so $n-1$ is not, leaving any valid subset of $\{1, \ldots, n-2\}$. That's the Fibonacci recurrence.

Recurrence. Take $n \geq 3$ and split the valid subsets S of $\{1, \ldots, n\}$:

- $n \notin S$: then S is any valid subset of $\{1, \ldots, n-1\}$, giving a_{n-1}.
- $n \in S$: then $n-1 \notin S$, and $S \setminus \{n\}$ is any valid subset of $\{1, \ldots, n-2\}$, giving a_{n-2}.

So $a_n = a_{n-1} + a_{n-2}$.

Initial values. $a_1 = 2$ ($\varnothing$, $\{1\}$) $= F_3$, and $a_2 = 3$ ($\varnothing$, $\{1\}$, $\{2\}$) $= F_4$.

Induction. a_n and F_{n+2} satisfy the same recurrence with the same first two values, so they are equal for all n. $\blacksquare$

Question 8.2

regions

n lines are drawn in the plane, no two parallel and no three through a point. Into how many regions do they divide the plane?

Hint: When you add the n-th line, how many pieces is it cut into by the others, and what does each piece do?

Solution 8.2

regions

n lines are drawn in the plane, no two parallel and no three through a point. Into how many regions do they divide the plane?

Key idea. The new line meets the $n-1$ old lines at $n-1$ distinct points, so it is cut into n pieces. Each piece splits one existing region in two.

Let R_n be the number of regions; $R_0 = 1$. Add the n-th line to a configuration of $n-1$ lines. It is not parallel to any of them, so it meets each exactly once, and these $n-1$ points are distinct (no three lines concurrent). They cut the new line into n pieces (two rays and $n-2$ segments). Each piece lies inside one old region and divides it into two. So

$$R_n = R_{n-1} + n \quad\Longrightarrow\quad R_n = 1 + (1 + 2 + \cdots + n) = 1 + \frac{n(n+1)}{2}.$$

Check: $R_1 = 2$, $R_2 = 4$, $R_3 = 7$. ✓

Answer: $\dfrac{n^2 + n + 2}{2}$ regions.

Question 8.3

inductive construction

Prove that for every $n \geq 1$, a $2^n \times 2^n$ board with any one square removed can be tiled by L-shaped trominoes (three squares forming an L).

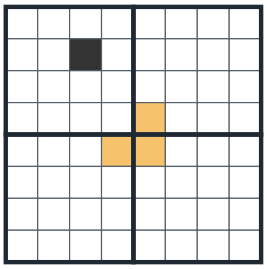
Hint: Split the board into four $2^{n-1} \times 2^{n-1}$ quarters. One has the hole; place one tromino in the centre to create a hole in each of the others.

Solution 8.3

inductive construction

Prove that for every $n \geq 1$, a $2^n \times 2^n$ board with any one square removed can be tiled by L-shaped trominoes (three squares forming an L).

Key idea. Induction on n. Split into four quarters; the removed square is in one of them. One L-tromino at the centre covers one corner square of each of the other three quarters, so each quarter now has exactly one missing square, and induction applies.



Hole (black) in one quarter; the central tromino (orange) makes a "hole" in each other quarter.

Base case $n = 1$. A 2×2 board minus one square is exactly an L-tromino.

Inductive step. Assume the result for $2^{n-1} \times 2^{n-1}$ boards. Split the $2^n \times 2^n$ board into four $2^{n-1} \times 2^{n-1}$ quarters. The removed square lies in exactly one quarter. Place one L-tromino on the three squares at the centre of the board that belong to the other three quarters; they form an L. Now each quarter is a $2^{n-1} \times 2^{n-1}$ board with exactly one square missing or covered, and by the inductive hypothesis each can be tiled. Together with the central tromino, this tiles the board. ■

Question 8.4

strong induction

Using the Fibonacci numbers $1, 2, 3, 5, 8, 13, \ldots$ (each at most once), prove that every positive integer can be written as a sum of distinct Fibonacci numbers, no two of which are consecutive in this list. (Zeckendorf's theorem, existence part.)

Hint: Greedy: subtract the largest Fibonacci number $\le n$. Show the remainder is less than the previous Fibonacci number.

Solution 8.4

strong induction

Using the Fibonacci numbers $1, 2, 3, 5, 8, 13, \ldots$ (each at most once), prove that every positive integer can be written as a sum of distinct Fibonacci numbers, no two of which are consecutive in this list. (Zeckendorf's theorem, existence part.)

Key idea. Greedy plus strong induction. If $f_k \le n < f_{k+1}$, then $n - f_k < f_{k+1} - f_k = f_{k-1}$, so the representation of $n - f_k$ uses only terms up to f_{k-2}, which are not adjacent to f_k.

Write the list as $f_1 = 1$, $f_2 = 2$, and $f_{k+1} = f_k + f_{k-1}$. We use strong induction on n. For $n = 1$: $1 = f_1$.

Let $n \ge 2$ and assume the claim for all smaller positive integers. Let f_k be the largest list term $\le n$, so $f_k \le n < f_{k+1}$. If $n = f_k$, we're done. Otherwise $m = n - f_k$ satisfies $1 \le m < f_{k+1} - f_k = f_{k-1}$.

By the hypothesis, m is a sum of distinct non-consecutive terms, each $\le m < f_{k-1}$, so each is at most f_{k-2}. Adding f_k gives a representation of n: all terms are distinct, and f_k is not consecutive to any of them (the largest is at most f_{k-2}). $\blacksquare$

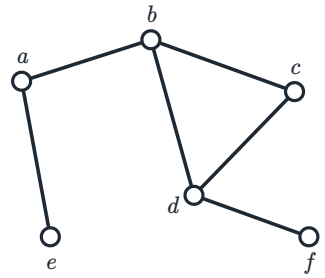
Example. $100 = 89 + 11 = 89 + 8 + 3$.

9 Graph theory

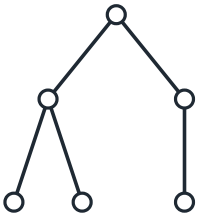
Strip away the story: dots and lines.

A huge number of olympiad problems are about things connected to other things: people and friendships, towns and roads, teams and matches. A **graph** is the language for all of them. Learning the basic vocabulary and a few facts lets you strip the story away and see the structure.

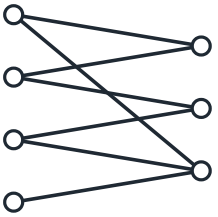
Vocabulary



a graph with 6 vertices and 6 edges



a tree



a bipartite graph

- A **graph** is a set of **vertices** (dots), some pairs of which are joined by **edges** (lines). In the first picture, b and d are **adjacent** (joined), a and c are not.
- The **degree** of a vertex is the number of edges at it. Above, $\deg b = 3$, $\deg e = 1$. By the handshake lemma (Section 7), $\sum \deg v = 2|E|$: here $2+3+2+3+1+1 = 12 = 2 \times 6$.
- A **path** is a sequence of distinct vertices, each joined to the next (e, a, b, d, f). A **cycle** is a path that returns to its start (b, c, d, b).
- A graph is **connected** if any two vertices are joined by a path.
- A **tree** is a connected graph with no cycles. A **leaf** is a vertex of degree 1.
- The **complete graph** K_n has n vertices and all $\binom{n}{2}$ possible edges.
- A graph is **bipartite** if its vertices can be split into two groups with every edge going between the groups (equivalently, it can be coloured with 2 colours so that each edge joins different colours).

Key fact: facts worth knowing

- **Handshake lemma:** $\sum_v \deg v = 2|E|$, so the number of odd-degree vertices is even.
- **Trees:** every tree on n vertices has exactly $n-1$ edges (Question 9.2), and every tree with at least 2 vertices has a leaf.
- **Bipartite:** a graph is bipartite if and only if it has no cycle of odd length.
- **Euler:** a connected graph has a closed walk using every edge exactly once if and only if every vertex has even degree.

Worked example 9.1 two vertices of the same degree

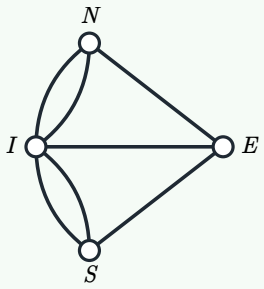
> Prove that at any party with at least 2 people, two people have the same number of friends at the party.

Model. People are vertices, friendships are edges. With n people, each degree is one of $0, 1, \ldots, n-1$: that's n possible values for n people, so pigeonhole doesn't immediately work.

The extra observation. The values 0 and $n-1$ can't *both* occur: a person of degree $n-1$ is friends with everyone, so nobody has degree 0. So the degrees take at most $n-1$ different values, and by pigeonhole two of the n people share a degree. ■

Worked example 9.2 the bridges of Königsberg

> A city has four land areas joined by seven bridges, as in the picture (land areas are vertices, bridges are edges). Can you take a walk that crosses every bridge exactly once and returns to the start?



Each time a closed walk passes through a vertex, it uses one edge to arrive and one to leave. So every vertex is met by an **even** number of the walk's edges. A walk using every edge exactly once would therefore need every degree to be even. Here the degrees are $\deg N = 3$, $\deg S = 3$, $\deg E = 3$, $\deg I = 5$: all odd. So it's impossible.

(Even an open walk, ending somewhere else, needs all but at most two degrees to be even. So that's impossible too.)

Method: model it as a graph

- Decide what the vertices and edges are. Draw the small cases.
- Think about **degrees** (handshake lemma, pigeonhole on degrees).
- Think about **longest paths** (extremal principle: the end of a longest path has all its neighbours on the path).
- **Colour** the edges (for "friends or strangers" problems) or the vertices (for bipartite arguments).
- **Count edges two ways** (Section 7).

Common trap

Be careful about which graph you mean. "Friends" and "strangers" are two different graphs on the same vertices (complements of each other). In Ramsey-type problems it's clearer to think of *one* complete graph whose edges are coloured in two colours.

Quick check

1. A graph has 7 vertices with degrees $1, 2, 2, 3, 3, 4, 4$. Is that possible?
2. How many edges does K_{10} have?
3. A tree has 4 vertices of degree 3 and all other vertices are leaves. How many leaves are there?

Answers: **1.** No: the degree sum 19 is odd. · **2.** 45 · **3.** 9 (if ℓ leaves: $1 + 4(2 - \ell) + 12 = \ell$)

Now practise

Each question below has its own page with space to work and a hint printed upside down at the bottom. The full solution is on the following page.

9.1 Ramsey $R(3,3) = 6$

9.2 trees

9.3 Mantel's theorem

Question 9.1

Ramsey $R(3,3) = 6$

Prove that among any 6 people there are either 3 mutual friends or 3 mutual strangers. Show this can fail for 5 people.

Hint: Pick one person. Of the 5 others, at least 3 have the same relationship to them.

Solution 9.1

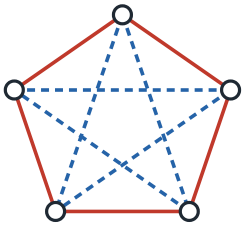
Prove that among any 6 people there are either 3 mutual friends or 3 mutual strangers. Show this can fail for 5 people.

Key idea. Colour each pair red (friends) or blue (strangers). One person has 5 edges, so at least 3 have the same colour, say red. Among those 3 people, either some edge is red, completing a red triangle, or all are blue.

Colour each pair of people red (friends) or blue (strangers). Fix a person v. Of the 5 edges at v, at least $\lceil 5/2 \rceil = 3$ have the same colour; say va, vb, vc are red (the blue case is the same with colours swapped).

- If any of ab, bc, ca is red, say ab, then v, a, b are mutual friends.
- Otherwise ab, bc, ca are all blue, and a, b, c are mutual strangers.

Five is not enough. Seat 5 people round a table; neighbours are friends (red, solid), everyone else strangers (blue, dashed). Red edges form a 5-cycle and blue edges the "pentagram" 5-cycle; neither contains a triangle. ■



Question 9.2

trees

Prove that every tree with $n \geq 1$ vertices has exactly $n - 1$ edges.

Hint: First show a tree with at least 2 vertices has a vertex of degree 1 (consider a longest path). Then remove it and use induction.

Solution 9.2

Prove that every tree with $n \geq 1$ vertices has exactly $n - 1$ edges.

Key idea. An endpoint of a longest path must be a leaf: another neighbour would give a cycle or a longer path. Removing a leaf leaves a smaller tree with one fewer edge.

Lemma: a tree with $n \geq 2$ vertices has a leaf. The tree is connected with at least 2 vertices, so it has an edge. Take a longest path $v_0 v_1 \dots v_\ell$ ($\ell \geq 1$). If v_0 had a neighbour $w \neq v_1$: if w is on the path, that gives a cycle; if not, $w v_0 \dots v_\ell$ is a longer path. Both are impossible, so $\deg v_0 = 1$.

Induction on n. $n = 1$: no edges. For $n \geq 2$, remove a leaf v and its edge. What remains is still connected (no path between two other vertices passes through a degree-1 vertex) and has no cycles, so it's a tree on $n - 1$ vertices. By induction it has $n - 2$ edges, so the original has $n - 1$. ■

Question 9.3

Prove that a graph with n vertices and more than $\dfrac{n^2}{4}$ edges contains a triangle. Show the bound is sharp.

Hint: If there's no triangle, the endpoints of an edge uv have no common neighbour, so $\deg u + \deg v \le n$. Sum over edges, then use Cauchy–Schwarz or QM–AM.

Solution 9.3

Prove that a graph with n vertices and more than $\dfrac{n^2}{4}$ edges contains a triangle. Show the bound is sharp.

Key idea. Triangle-free means each edge's endpoints share no neighbour, so $\deg u + \deg v \le n$. Summing over edges gives $\sum \deg^2 \le nE$. Cauchy–Schwarz gives $\sum \deg^2 \ge (2E)^2/n$. Together, $E \le n^2/4$.

Suppose the graph is triangle-free, with E edges and degrees d_v.

Edge bound. For an edge uv, no vertex is adjacent to both u and v (that would make a triangle). The neighbourhoods of u and v are disjoint subsets of the n vertices, so $d_u + d_v \le n$.

Sum over edges. $\sum_{uv \in E} (d_u + d_v) = \sum_v d_v^2$, since vertex v appears in d_v edges. So $\sum_v d_v^2 \le nE$.

Cauchy–Schwarz. $\sum_v d_v^2 \ge \frac{1}{n}\left(\sum_v d_v\right)^2 = \frac{(2E)^2}{n}$. Hence $\frac{4E^2}{n} \le nE$, i.e. $E \le \frac{n^2}{4}$. So more than $\frac{n^2}{4}$ edges forces a triangle.

Sharp. For even n, split the vertices into two halves of $\frac{n}{2}$ and join every vertex in one half to every vertex in the other. This has $\frac{n^2}{4}$ edges, and no triangle, since any triangle would need two vertices in the same half. ■

10 Games

Who wins, and how?

Game problems ask: in a two-player game with given rules, who wins with perfect play, and how? An answer needs a **strategy** for the winner and a proof that it works *whatever the opponent does*. Three methods cover almost every olympiad game.

Throughout, the games are finite (they always end), have no draws, and both players see everything. For such games one of the two players always has a winning strategy; the question is which.

Method 1: winning and losing positions

Call a position **P** (a **P**revious-player win: the player *to move* loses) or **N** (a **N**ext-player win: the player to move wins). The rules:

Key fact: P and N positions

- A position where the player to move has **no moves** (and so has lost) is P.
- A position is **N** if **some** move leads to a P position (move there, and your opponent is stuck in a losing position).
- A position is **P** if **every** move leads to an N position (whatever you do, your opponent is now winning).

The winning strategy is simple: **always move to a P position.**

Worked example 10.1 a subtraction game by table

> A pile has n stones. Players alternately remove 1, 3 or 4 stones. Whoever takes the last stone wins. For which n does the second player win?

Build the table from 0 upwards. $n = 0$: no move, so P. $n = 1$: can move to 0 (P), so N. $n = 2$: the only move is to 1 (N), so P. $n = 3$: move to 0, N. $n = 4$: move to 0, N. $n = 5$: can reach 4 or 2, and 2 is P, so N. $n = 6$: can reach 5, 3, 2; 2 is P, so N. $n = 7$: reaches 6, 4, 3, all N, so P. Continuing:

n	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
type	**P**	N	**P**	N	N	N	N	**P**	N	**P**	N	N	N	N	**P**	N	**P**

Guess. The P positions are $n \equiv 0$ or $2 \pmod 7$.

Prove. We check the two rules. (1) From a P position ($n \equiv 0, 2$) every move goes to an N position: removing $1, 3, 4$ from $n \equiv 0$ gives $6, 4, 3$, and from $n \equiv 2$ gives $1, 6, 5 \pmod 7$; none is 0 or 2. (2) From every other position some move goes to P: from 1, remove 1 (to 0); from 3, remove 1 (to 2) or 3; from 4, remove 4 (to 0); from 5, remove 3 (to 2); from 6, remove 4 (to 2). So the second player wins exactly when $n \equiv 0$ or $2 \pmod 7$.

Common trap

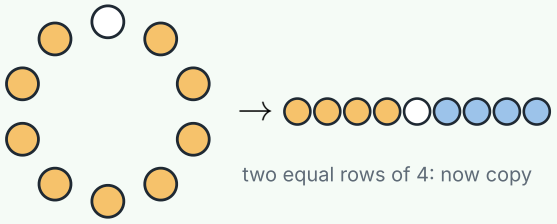
A pattern in a table is only a guess. The proof is checking the two rules (every move from P goes to N; some move from N goes to P) for **all** positions, usually using the pattern's form (here, residues mod 7).

Method 2: symmetry (copy the opponent)

If you can reach a **symmetric** position, you can then answer each opponent move with its mirror image. The opponent always moves first in each symmetric pair, so if they can move, so can you, and you never run out of moves first. The one thing to check: **the mirror move is always legal**.

Worked example 10.2 coins in a circle

> $n \geq 3$ coins are arranged in a circle. Players alternately take either one coin, or two coins that are next to each other. Whoever takes the last coin wins. Who wins?



After the first player takes one coin, the second takes the middle one.

The second player wins. After the first player's move, the remaining coins form a single row (the circle has been broken). If that row has at most 2 coins, take them all and win. Otherwise take 1 or 2 coins from the **exact middle** of the row (1 if the row has odd length, 2 if even), leaving two equal rows. From now on, whatever the first player takes from one row, take the same coins from the matching position in the other row.

Legality. After each of the second player's moves, the position is two identical copies. The first player's move affects only one copy (coins taken together are adjacent, so they're in the same row), and the matching coins in the other copy are still there, and still adjacent. So the second player can always move, and therefore the first player is the one who eventually can't.

Method 3: strategy stealing

Sometimes you can prove the first player wins without knowing how. Suppose, for contradiction, that the second player has a winning strategy. If the first player can make some move that "does no harm" (a move that could just as well have been part of any later position), the first player can make it and then pretend to be the second player, following the second player's strategy. That would make *both* players winners: a contradiction. Question 10.3 is the classic example.

Method: three standard strategies

- **P/N positions:** compute small cases, guess the pattern, then prove: from P every move goes to N, and from N some move goes to P.
- **Symmetry:** make the position symmetric (often using the centre or the middle), then copy. Check the copied move is always legal.
- **Strategy stealing:** if an extra move can never hurt, the first player wins (but we may not know how).
- Always state **who** wins and **describe the strategy**; then prove it works against every reply.

Quick check

1. Players alternately remove 1 or 2 stones from a pile; whoever takes the last stone wins. Which n are P positions?
2. Same game, but whoever takes the last stone *loses*. Which $n \geq 1$ are P positions?
3. In the 1, 3, 4 game above, who wins with $n = 100$?

Answers: **1.** multiples of 3 · **2.** $n \equiv 1 \pmod 3$ · **3.** $100 \equiv 2 \pmod 7$: the second player

Now practise

Each question below has its own page with space to work and a hint printed upside down at the bottom. The full solution is on the following page.

10.1 P-positions

10.2 symmetry

10.3 strategy stealing: Chomp

10.4 Nim

Question 10.1

P-positions

There are n stones in a pile. Two players alternately remove 1, 2 or 3 stones. The player who takes the last stone wins. For which n does the first player have a winning strategy?

Hint: Compute who wins for $n = 0, 1, \ldots, 8$.

Solution 10.1

P-positions

There are n stones in a pile. Two players alternately remove 1, 2 or 3 stones. The player who takes the last stone wins. For which n does the first player have a winning strategy?

Key idea. Small cases suggest the losing positions are the multiples of 4. Prove it: from a multiple of 4 every move leaves a non-multiple, and from a non-multiple you can always move to a multiple of 4.

n	0	1	2	3	4	5	6	7	8	9	10	11	12
type	**P**	N	N	N	**P**	N	N	N	**P**	N	N	N	**P**

Claim: the player to move loses if and only if $4 \mid n$.

- $n = 0$: the player to move has no stones to take, because the opponent took the last one, so they have lost.
- From $n \equiv 0 \pmod 4$, $n > 0$: removing 1, 2, 3 leaves $n-1, n-2, n-3$, none divisible by 4.
- From $n \not\equiv 0$: remove $r = n \bmod 4 \in \{1, 2, 3\}$ stones, leaving a multiple of 4.

So a player facing a non-multiple of 4 can always hand the opponent a multiple of 4, and the opponent can never return one. The pile shrinks, so eventually the opponent faces 0 and has lost.

Answer: The first player wins if and only if n is not a multiple of 4.

Question 10.2

symmetry

Two players take turns placing identical circular coins on a circular table. Coins must lie entirely on the table and must not overlap. The player who cannot move loses. Who wins?

Hint: The first player can use the centre of the table.

Solution 10.2

symmetry

Two players take turns placing identical circular coins on a circular table. Coins must lie entirely on the table and must not overlap. The player who cannot move loses. Who wins?

Key idea. The first player puts a coin exactly in the centre, then answers every move with its reflection through the centre. The position stays symmetric, so the reflected move is always legal.

Strategy for the first player. First, place a coin with its centre at the table's centre O. Afterwards, whenever the second player places a coin C, place C', the reflection of C through O.

Legality. After each of the first player's moves, the arrangement is symmetric about O. Suppose the second player has just legally placed C (radius r).

- C' is on the table, since the table is symmetric about O.
- C' doesn't overlap C: C doesn't overlap the central coin, so the centre of C is at distance at least $2r$ from O. Then the centres of C and C' are at distance at least $4r$.
- C' doesn't overlap any earlier coin D: otherwise C would overlap D', which is also on the table by symmetry. That would make C illegal.

So the first player can always move. The game ends (only finitely many coins fit), so it is the second player who eventually cannot move.

Answer: The first player wins.

Question 10.3

strategy stealing: Chomp

A chocolate bar is an $m \times n$ grid of squares ($mn > 1$); the top-left square is poisoned. Players alternately choose a remaining square and eat it together with all squares below it and to its right. Whoever eats the poisoned square loses. Prove that the first player has a winning strategy.

Hint: Suppose the second player has a winning strategy. What if the first player starts by eating only the bottom-right square?

Solution 10.3

strategy stealing: Chomp

A chocolate bar is an $m \times n$ grid of squares ($mn > 1$); the top-left square is poisoned. Players alternately choose a remaining square and eat it together with all squares below it and to its right. Whoever eats the poisoned square loses. Prove that the first player has a winning strategy.

Key idea. If the second player could win, they'd have a winning reply to "eat the bottom-right square". But that reply removes the bottom-right square anyway, so the first player could have played it as their first move, and stolen the win.

The game is finite with no draws, so one player has a winning strategy. Suppose it is the second player.

Let the first player eat only the bottom-right square s (allowed since $mn > 1$, so s is not the poisoned square). The second player's strategy gives a reply, eating some square x, after which the second player (now "the player who just moved") wins with correct play.

Eating x removes every square below and to the right of x, and s is one of these. So the position after "s then x" is the same as after just "x". Hence the first player could have eaten x on move one, reaching that same winning position as the player who just moved, and would then win by following the second player's strategy. This contradicts the assumption that the second player has a winning strategy. So the first player does. ■

Question 10.4

There are several piles of stones, of sizes $h_1, \dots, h_k$. Players alternately choose a pile and remove any positive number of stones from it. Whoever takes the last stone wins. Prove that the player to move loses exactly when $h_1 \oplus h_2 \oplus \cdots \oplus h_k = 0$, where $\oplus$ is binary XOR (add in binary without carrying).

Hint: Two facts are needed. From XOR 0, every move gives XOR $\neq 0$. From XOR $s \neq 0$, look at the highest binary digit of s and find a pile that has a 1 there.

Solution 10.4

There are several piles of stones, of sizes $h_1, \ldots, h_k$. Players alternately choose a pile and remove any positive number of stones from it. Whoever takes the last stone wins. Prove that the player to move loses exactly when $h_1 \oplus h_2 \oplus \cdots \oplus h_k = 0$, where $\oplus$ is binary XOR (add in binary without carrying).

Key idea. Positions with XOR 0 are the losing ones. Changing one pile always changes the XOR, so from 0 you must leave non-zero. From non-zero s, choose a pile h with a 1 in the top bit of s and reduce it to $h \oplus s < h$, which makes the XOR 0.

Let $s = h_1 \oplus \cdots \oplus h_k$. (Example: piles $3, 5, 6$ give $011 \oplus 101 \oplus 110 = 000$, a losing position for the player to move.)

From $s = 0$, every move gives $s \neq 0$. A move changes one pile h_i to $h_i' \neq h_i$. The new XOR is $s \oplus h_i \oplus h_i' = h_i \oplus h_i' \neq 0$.

From $s \neq 0$, some move gives $s = 0$. Let 2^d be the highest power of 2 in the binary expansion of s. Since the 2^d digit of s is 1, an odd number of piles have a 1 in that digit; pick one, h_i. Let $h_i' = h_i \oplus s$. In binary, h_i' agrees with h_i above digit d, and has 0 where h_i has 1 at digit d. So $h_i' < h_i$, and the move "reduce pile i to h_i'" is legal. The new XOR is $s \oplus h_i \oplus h_i' = s \oplus s = 0$.

Conclusion. The final position (no stones) has XOR 0, and the player to move there has lost. A player facing $s \neq 0$ can always hand over $s = 0$, and the opponent can never hand back $s = 0$. Every move removes stones, so the game ends, with the player facing $s = 0$ losing. ■

11 Extremal problems: bound + construction

Two halves, both required.

"Find the largest (or smallest) possible…" problems are the most common type at BMO1, and the most commonly half-solved. A complete answer always has **two independent parts**, and both need proof.

Key fact: the two halves of every "find the max"

To show the maximum is M:

1. **Bound:** prove that *every* valid configuration has size at most M.
2. **Construction:** exhibit *one* valid configuration of size exactly M, and check it is valid.

The bound alone doesn't show M is reached; the construction alone doesn't show you can't do better. For a minimum, swap the roles: prove every configuration has size at least M, and give one with exactly M.

Worked example 11.1 rooks: the simplest case

> What is the largest number of rooks that can be placed on an 8×8 board so that no two attack each other (a rook attacks along its row and column)?

Bound. Two rooks in the same row attack each other, so each of the 8 rows contains at most one rook: at most 8.

Construction. Put rooks on the main diagonal, squares $(1,1), (2,2), \ldots, (8,8)$. They are in different rows and different columns, so none attack each other.

Answer: 8. Notice the shape of the bound: we **split the board into 8 pieces** (rows), each of which can hold at most one rook. This "partition into pieces with at most one each" is pigeonhole in disguise, and it's the most common way to prove a bound.

Worked example 11.2 no two consecutive

> What is the largest subset of $\{1, 2, \ldots, 20\}$ containing no two consecutive numbers?

Small cases first. From $\{1,2,3,4\}$ the best is 2, e.g. $\{1,3\}$; from $\{1,\ldots,6\}$, 3. It looks like half.

Bound. Split $\{1,\ldots,20\}$ into the 10 pairs $\{1,2\}, \{3,4\}, \ldots, \{19,20\}$. Each pair is two consecutive numbers, so a valid subset contains at most one from each pair: at most 10 elements.

Construction. The odd numbers $\{1,3,5,\ldots,19\}$: 10 numbers, any two differ by at least 2.

Answer: 10.

Worked example 11.3 a minimum problem

> What is the smallest number of squares of an 8×8 board that must be marked so that every domino placed on the board (covering two adjacent squares) covers at least one marked square?

Bound (at least 32). The board can be tiled by 32 non-overlapping dominoes (e.g. 4 in each row). Each of these dominoes must cover a marked square, and they don't overlap, so at least 32 squares are marked.

Construction (32 works). Mark all 32 black squares of the chessboard colouring. Any domino covers one black and one white square, so it covers a marked square.

Answer: 32. For a minimum, the bound comes from finding many *disjoint* things that each need their own mark.

Where do bounds and constructions come from?

- **Bounds:** pigeonhole (partition into parts that each contain at most one element); double counting (Section 7); invariants and colourings; sometimes an averaging argument.
- **Constructions:** look at the small cases. The best examples for $n = 2, 3, 4$ usually show a pattern (every other square, all the odd numbers, the top half). Often the extremal configuration is very regular or symmetric.
- The two halves guide each other. If you can't push a construction past M, ask *what's stopping you*: that obstacle is the bound. If your bound proof has slack, a better construction may exist.

Common trap

A solution that only gives an example, or only proves a bound, is incomplete. At BMO1, this typically scores 2–4 out of 10. Label the two parts clearly ("Bound.", "Construction.") so the marker sees you've done both, and in the construction, actually **verify** that your example satisfies the conditions.

Quick check

1. What is the largest number of bishops on a 4×4 board with no two attacking? (Try it.)
2. What is the largest subset of $\{1, \ldots, 10\}$ with no two elements differing by exactly 5?
3. Fewest squares to mark on a 4×4 board so every 1×2 domino covers a marked one?

Answers: **1.** 9 (in general $n^2 - 2n$) · **2.** 5: pairs $\{k, k+5\}$; take 1, 2, 3, 4, 5 · **3.** 8

Now practise

Each question below has its own page with space to work and a hint printed upside down at the bottom. The full solution is on the following page.

11.1 kings

11.2 no divisibility

11.3 residues

Question 11.1

kings

What is the maximum number of kings that can be placed on an 8×8 chessboard so that no two attack each other? (A king attacks the up to eight squares adjacent to it, including diagonally.)

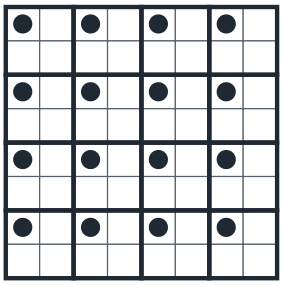
Hint: Cut the board into 2×2 blocks.

Solution 11.1

kings

What is the maximum number of kings that can be placed on an 8×8 chessboard so that no two attack each other? (A king attacks the up to eight squares adjacent to it, including diagonally.)

Key idea. Any two squares of a 2×2 block are adjacent, so each block holds at most one king. That gives 16, and putting one king in the same corner of every block achieves it.



Bound. Divide the board into 16 blocks of size 2×2. Any two squares in one block are adjacent (horizontally, vertically or diagonally), so each block contains at most one king. Hence at most 16 kings.

Construction. Place a king in the top-left square of every block, i.e. on all squares with odd row and odd column numbers. Any two of these are at least 2 apart in row or in column, so none attack each other. That's 16 kings.

Answer: 16.

Question 11.2

no divisibility

What is the largest size of a subset of $\{1, 2, \ldots, 2n\}$ in which no element divides another?

Hint: Question 3.1(b) gives the bound. For the construction, take the largest numbers.

Solution 11.2

no divisibility

What is the largest size of a subset of $\{1, 2, \ldots, 2n\}$ in which no element divides another?

Key idea. Bound: grouping by odd part gives n classes, and each class holds at most one element. Construction: $\{n+1, \ldots, 2n\}$, where the smallest multiple of any element is already too big.

Bound. Group the numbers by odd part: $x = 2^k m$, $m \in \{1, 3, \ldots, 2n-1\}$, giving n groups. Two numbers in the same group differ by a power of 2, so one divides the other. So a valid subset has at most one element per group, at most n in total.

Construction. $\{n+1, n+2, \ldots, 2n\}$ has n elements. If $a \mid b$ with $a < b$ in this set, then $b \geq 2a \geq 2n+2 > 2n$, which is impossible.

Answer: n.

Question 11.3

residues

What is the largest size of a subset of $\{1, 2, \ldots, 50\}$ in which no two distinct elements have a sum divisible by 7?

Hint: Sort the numbers by remainder mod 7. Which remainder classes can't both be used? How many elements of remainder 0 can there be?

Solution 11.3

residues

What is the largest size of a subset of $\{1, 2, \ldots, 50\}$ in which no two distinct elements have a sum divisible by 7?

Key idea. Two numbers sum to a multiple of 7 iff their remainders are r and $7 - r$, or both 0. So from each pair of classes $\{1, 6\}, \{2, 5\}, \{3, 4\}$ we use at most one whole class, and at most one multiple of 7.

Class sizes. Among $1, \ldots, 50$: remainder 1 occurs 8 times $(1, 8, \ldots, 50)$, and each of the remainders $0, 2, 3, 4, 5, 6$ occurs 7 times.

Bound. Let S be valid. If $x, y \in S$ had remainders r and $7 - r$ ($r \in \{1, 2, 3\}$), then $7 \mid x + y$. So for each pair of classes $\{1, 6\}, \{2, 5\}, \{3, 4\}$, S meets only one class. Also S contains at most one multiple of 7. Hence

$$|S| \le \max(8, 7) + \max(7, 7) + \max(7, 7) + 1 = 23.$$

Construction. Take all numbers with remainder 1, 2 or 3, plus the number 7: $8 + 7 + 7 + 1 = 23$ elements. Remainders of sums of two distinct elements lie in $\{1 + 1, 1 + 2, \ldots, 3 + 3\} \cup \{0 + 1, 0 + 2, 0 + 3\} = \{2, 3, 4, 5, 6\} \cup \{1, 2, 3\}$, never 0. And there is only one multiple of 7.

Answer: 23.